The British Accounting Review xxx (xxxx) xxx

Contents lists available at ScienceDirect

The British Accounting Review

journal homepage: www.elsevier.com/locate/bar

Beware of false prophets: Cybersecurity risk and strategic voluntary disclosure☆

Hong Vo [a,b], Man Duy Pham [c,d,*]

[a] International University, Quarter 6, Linh Trung Ward, Thu Duc City, Ho Chi Minh City, Viet Nam
[b] Vietnam National University, Ho Chi Minh City, Viet Nam
[c] University of Auckland Business School, University of Auckland, Auckland, New Zealand
[d] Department of Business, Swinburne Vietnam, FPT University, Ho Chi Minh City, Vietnam

ARTICLE INFO

JEL classification:
D81
D82
G30
G32
M41

Keywords:
Management forecasts
Strategic disclosure
Cybersecurity risk
Managerial opportunism

ABSTRACT

This study explores how cybersecurity risk influences voluntary disclosure, with a particular focus on management forecasts. We find that managers in firms with higher ex-ante cyber risk issue more financial forecasts. This positive effect is largely due to increased information demand from market participants when confronted with elevated cybersecurity concerns. Further analyses reveal that management forecasts in response to cyber risk, while conveying more positive and long-term information, exhibit lower precision and accuracy. Overall, we interpret these findings as managers strategically using optimistic disclosures, even if these disclosures' quality is subpar, to enhance market valuation when their firm is exposed to cybersecurity risk.

> "The distinction between a commitment and a voluntary disclosure is that the former is a decision by the firm about what it will disclose before it knows the content of information (i.e., ex-ante), whereas the latter is a decision by the firm made after it observes the content (i.e., ex-post) …"
>
> Leuz and Verrecchia [2000, p.94]

1. Introduction

The past decades have witnessed a transformative surge in information technology, fueling the exponential growth of cyberspace, digital transformation, and global interconnectivity. While undeniably positive, propelling the global economy forward and unlocking new business opportunities, these changes pose their own set of risks. As companies' dependence on digital systems intensifies, so do

☆ The authors thank seminar participants at the University of Auckland, International University (Vietnam National University HCMC), and Monash University (Malaysia) for their helpful comments. Part of this study was completed while Pham was affiliated with Monash University. Pham acknowledges research funding from the University of Auckland. All errors are ours.

* Corresponding author. Man Duy (Marty) Pham – Department of Accounting and Finance, University of Auckland Business School, University of Auckland, 12 Grafton Rd, Auckland, New Zealand.
E-mail address: man.pham@auckland.ac.nz (M.D. Pham).

https://doi.org/10.1016/j.bar.2025.101578
Received 29 March 2024; Received in revised form 23 October 2024; Accepted 17 January 2025
Available online 21 January 2025
0890-8389/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Please cite this article as: Hong Vo, Man Duy Pham, *The British Accounting Review,* https://doi.org/10.1016/j.bar.2025.101578

the systematic and evolving concerns about corporate cybersecurity.[1] Recent successful cyberattacks have demonstrated the devastating consequences of these threats, leading to data breaches, operational disruptions, and reputational damage for targeted companies. As such, stakeholders, including shareholders, customers, employees, and suppliers, are increasingly aware of these risks, making cybersecurity a top concern and challenge for managers worldwide (PwC, 2018; World Economic Forum, 2020).[2]

The mounting prevalence of cybersecurity has spurred significant research into the value implications of cyber risk, both for individual firms and the broader financial market (e.g., Berkman, Jona, Lee, & Soderstrom, 2018; Florackis, Louca, Michaely, & Weber, 2023; Jiang, Khanna, Yang, & Zhou, 2024; Lattanzio & Ma, 2023).[3] Yet, a critical gap remains in our understanding of how managers respond to and interact with shareholders when their firm has a higher exposure to cybersecurity problems. This study aims to bridge this gap by examining managerial response – in the form of voluntary disclosure – to *ex-ante* cybersecurity risk. We focus specifically on the issuance of management forecasts, one of the most important and widely studied forms of voluntary financial disclosure (Hirst, Koonce, & Venkataraman, 2008). Within the context of this study, we propose that the relationship between cybersecurity risk and management forecasts is multifaceted.

On the one hand, cybersecurity risk may discourage managers from issuing voluntary financial projections for two main reasons. First, cyber-related affairs impose substantial, yet often immeasurable, financial burdens on both firms and the financial market. For instance, reputational damage associated with cyber incidents, which is notoriously difficult to quantify, is a significant component of these losses (Aldasoro, Gambacorta, Giudici, & Leach, 2022; Kamiya, Kang, Kim, Milidonis, & Stulz, 2021; Makridis, 2021).[4] Moreover, the imperative to implement robust cybersecurity measures as part of risk management strategies in more exposed firms necessitates significant resource allocation, operational complexity, and innovative solutions (Dewan, Shi, & Gurbaxani, 2007; Wang & Alam, 2007). The confluence of these financial and operational uncertainties can hinder managers' ability to produce reliable and timely forecasts. Secondly, management forecasts raise the risk of exposing sensitive strategic information and expected future cash outlays to competitors, eroding a firm's competitive edge (Bamber & Cheon, 1998; Huang, Jennings, & Yu, 2017; Verrecchia, 1983). This risk is pertinent to firms with higher cybersecurity risk, as leaking proprietary information further jeopardizes their market position or, even worse, increases the likelihood of becoming a cyber-attack target (Ettredge, Guo, & Li, 2018). Under these circumstances, managers could refrain from providing forecasts to safeguard their firm's competitive position and personal credibility from inaccuracies or delays in disclosure (Blankespoor, deHaan, & Marinovic, 2020; Feng, Li, & McVay, 2009). We term this the *costly disclosure* hypothesis.

Nevertheless, the strategy of silence mentioned above is also costly because it contravenes a firm's commitment to transparent disclosure practices and exacerbates information asymmetry (Chen, Matsumoto, & Rajgopal, 2011; Houston, Lev, & Tucker, 2010). The non-compliance costs, as we reckon, are even higher for firms confronted with cyber risk, given that these entities are typically subject to more extensive disclosure requirements of cybersecurity management or breaches, as mandated by the SEC's CF Disclosure Guidance (Topic 2) issued in 2011.[5] Besides, cybersecurity concerns can incite stakeholder demands for greater information transparency.[6] Failure to fulfill these expectations could magnify adverse selection problems, where potential investors with limited knowledge of a company's cybersecurity vulnerabilities tend to undervalue its shares (Bertomeu, Marinovic, Terry, & Varas, 2022; Dye, 1985; Jung & Kwon, 1988). Therefore, rather than adopting a strategy of silence, exposure to cyber risk may urge managers to deliberately extend their forecasts to maintain stakeholder confidence and mitigate the adverse price impact of cyber risk. Consistent with this perspective, prior research shows that firms disclose more narrative information about cybersecurity following a data breach to mitigate investors' contagion effects (Jiang, Legoria, Reichelt, & Walton, 2022; Kelton & Pennington, 2020) or when facing higher

[1] In 2020, cybercrime inflicted a substantial cost on the global economy, estimated at nearly USD 1 trillion (McAfee, 2020). This figure represents a significant rise of over 50% compared to 2018. This surge coincides with a dramatic increase in malware attacks detected worldwide. Data reveals a staggering 3.5-fold increase in such attacks from 2019 to 2020, with the trend showing no signs of abating in 2021 and 2022 (BusinessWire, 2021). Furthermore, the ongoing war in Ukraine has exacerbated the cyber threat landscape. Phishing attacks targeting European and US businesses, linked to actors in Russia, have reportedly skyrocketed, highlighting the evolving and increasingly concerning nature of cybercrime (Violino, 2023).

[2] According to the PwC, 2018 *Global Investor Survey*, 41% of investors and analysts rank cybersecurity top of the business threat list. the survey can be accessible at: https://www.pwc.com/kz/en/pwc-news/what-new/investors-put-cybersecurity.html.

[3] [3] It is essential to distinguish this research stream, which focuses on how *ex-ante* cybersecurity risk exposure influences firm and market outcomes, from studies examining *ex-post* market consequences or firm responses to cyber incidents or attacks (e.g., Amir et al., 2018; Kamiya et al., 2021).

[4] As for an anecdotal evidence, Target's major data breach in 2013 costs the company $18.5 million in a multi-state settlement, over $100 million for upgrading security systems, and $10 million in a class-action lawsuit settlement. Beyond these direct and immediate costs, Target also suffered reputational damage and lost its customer trust, thereby affecting holiday sales. The company then reported a drop in profit of nearly 46% in the quarter following the breach.

[5] We acknowledge a potential bias in our empirical analysis due to the introduction of the SEC's 2011 disclosure rule. As our cybersecurity risk measure is derived from the textual information in 10-K filings between 1998 and 2019, the rule's implementation may have influenced our findings. To address this, we conduct a robustness check by examining pre- and post-2011 subsamples. Our results indicate that the baseline relationship remains qualitatively consistent.

[6] Anecdotal evidence suggests that investor concerns about cybersecurity have demonstrably augmented the demand for information transparency in financial reports, especially about the financial risks associated with cyber-related events, regardless of whether these events are negative (e.g., cyberattacks) or positive (e.g., cyber preparedness)"https://www.forbes.com/sites/forbestechcouncil/2022/11/08/coming-clean-why-cybersecurity-transparency-is-a-strength-not-a-weakness/?sh=58188f04be24","https://corpgov.law.harvard.edu/2023/08/17/how-to-prepare-for-cyber-disclosures-in-a-new-era-of-transparency.

cyber risk levels (Walton, Wheeler, Zhang, & Zhao, 2021). We refer to this view as the *market conditioning* hypothesis. In summary, whether the costly disclosure hypothesis, the market conditioning hypothesis, or a combination thereof determines the overall association between cybersecurity risk and management forecasts remains an empirical question.

To investigate this question, we utilize a sample of 51,548 U.S. firm-year observations from 1998 to 2019. The sample period is determined by the availability of the cybersecurity risk measure developed by Lattanzio and Ma (2023). This measure, constructed through textual analysis of firms' 10-K filings, is used as our main proxy for the level of cyber risk faced by each firm within the sample. To quantify voluntary disclosure, we employ the number of management forecasts issued in a given fiscal year obtained from the I/B/E/S Guidance database.

Our main analyses document a positive association between cybersecurity risk and voluntary disclosure. On average, we find that a one-standard-deviation increase in cybersecurity risk score is associated with an annual 8.54% increase in the frequency of management forecasts. This trend gets stronger when comparing firms whose cyber risk score is above the sample median: they produce 12.55% more forecasts than their counterparts with below-median scores. We subject these findings to a rigorous battery of robustness checks and identification strategies to alleviate potential endogeneity issues, and the results continue to hold. Overall, our baseline evidence sides with the *market conditioning* proposition.

To further elucidate this relationship, we perform subsample analyses categorized by firms' information demand levels. Results indicate that the positive impact of cybersecurity risk on management forecast frequency is more pronounced among firms with higher information demand. This suggests that managers are likely to employ their forecasts to meet stakeholders' increased demand for transparency in a volatile operating environment.

While the baseline results, on average, support a positive relationship between cybersecurity risk and management forecasts, a critical question emerges as to the quality and reliability of these forecasts. We propose two possible answers to this question. The first one is based solely on the *market conditioning* hypothesis that managers issue more forecasts in the goodwill of fostering transparency and alleviating investors' concerns about management competence or preparedness (Jiang et al., 2022; Walton et al., 2021). Under this scenario, forecasts issued in response to elevated cyber risk should be informative and predictive of future performance. On the contrary, another answer hinges on the co-existence of the *costly disclosure* hypothesis and the *market conditioning* hypothesis. Considering the uncertainties induced by cybersecurity problems and the concurrent pressure to provide information to stakeholders, managers may strategically decide to issue forecasts to influence market perceptions, potentially at the expense of forecast quality. If this is the case, management forecasts issued by high-cybersecurity-risk firms are less credible and represent a form of market distraction.[7]

Further empirical analyses provide support for the latter perspective. We find that managers in firms with a higher cybersecurity risk issue positive and long-run forecasts more frequently to manage investor relations.[8] However, their forecasts are less precise and less accurate. We interpret these findings as indicative of managers' strategic behavior in navigating the competing demands of increased transparency with deteriorated forecasting ability due to cybersecurity risk. More intriguingly, when we delve deeper into the market implications of these strategic forecasts, our results show that they are associated with a higher risk of future stock price crashes and are less trusted by investors, especially when preceded by a series of optimistic predictions. Hence, we argue that these forecasts, albeit seemingly innocuous, may signal managerial bad news hoarding and contribute to market mispricing. This evidence underpins strategic voluntary disclosures as a potential explanation for the positive association between cybersecurity risk and stock price crash risk, as documented in Florackis et al. (2023).

Having established evidence of our critical findings, we examine the heterogeneous effects of cyber risk on management forecasts across different firm-level characteristics. Our analyses reveal that strategic disclosures aimed at influencing market perception are less likely to occur when the associated costs outweigh the potential benefits. More specifically, we find that the amplifying effect of cybersecurity risk on management forecasts is weakened in more reputable firms, those with more complex information production processes, and those bearing higher proprietary costs of disclosure arising from product market competition or asset intangibility.

Finally, we investigate the effectiveness of external governance mechanisms in abating strategic managerial behavior in disclosure. The findings indicate that external monitoring of the market for corporate control (e.g., takeovers market) and market regulators can significantly counteract the effects of cybersecurity risk on disclosure strategies. In contrast, well-known internal governance mechanisms, such as shareholder litigation, corporate boards, and incentive contracts, appear less effective in this context.

This study contributes to the growing body of research exploring the economic consequences of *ex-ante* cybersecurity risk on both market and firm-level outcomes. For example, Berkman et al. (2018) demonstrate that stock market participants react favorably to corporate cyber disclosures within 10-K filings. Similarly, Florackis et al. (2023) find that cybersecurity risk is associated with higher stock returns and predicts a higher likelihood of price crashes. Jiang et al. (2024) add this link and document a cyber risk premium in stock pricing. In the corporate landscape, Lattanzio and Ma (2023) find that firms facing greater cybersecurity risk prioritize faster innovation cycles by filing for simpler but less valuable patents.

Our study deviates from the existing research by looking into how managers leverage voluntary financial disclosures to manage investor relations when their firms are exposed to higher cyber risk. Our findings suggest that voluntary forecasts can be strategically

[7] This perspective is congruent with an industry report by Koller, Relm, and Williams (2021), which suggests that earnings guidance may often serve as a distraction rather than a valuable tool for investors, as investors may assign less weight to such guidance than executives believe.

[8] Previous literature asserts that the use of long-run forecasts rather than short-run ones might help managers minimize the potential reputational loss when issuing low-quality forecasts since long-run predictions often give managers more opportunities to revise their forecasting errors (Tang et al., 2023) or to attribute these errors to unavoidable fluctuations over a longer period of time (Gong, Li, & Xie, 2009; Hutton & Stocken, 2021).

used as a market conditioning tool. By proactively disseminating financial forecasts, managers can present a positive view of their firm to reduce potential share price discounts arising from adverse selection problems in the presence of cybersecurity risk. However, it is important to acknowledge that these forecasts may not always be credible, potentially leading to market mispricing. Investors should, therefore, exercise caution and conduct a thorough analysis when making their investment decisions to mitigate misleading forecasts in the cyber age.

Furthermore, this study contributes to the ongoing discourse surrounding the impact of voluntary disclosure on firms' information environment. Existing literature suggests that such disclosures can help reduce information asymmetry and lower the cost of capital (Diamond & Verrecchia, 1991; Shroff, Sun, White, & Zhang, 2013). Having said that, Leuz and Verrecchia (2000) point out the discretionary nature of voluntary disclosures, which triggers concerns about the credibility and informativeness of these disclosures. Consistent with this view, empirical evidence indicates managers can opportunistically determine the level, timing, and quality of the information in voluntary disclosures to serve their self-interests (Baginski, Campbell, Hinson, & Koo, 2018; Brockman, Khurana, & Martin, 2008; Cheng, Luo, & Yue, 2013; Pae, Song, & Yi, 2016; Pham & Nguyen, 2024).

Our study substantiates these findings by providing evidence of managers in firms with higher cybersecurity risk opportunistically releasing more positive but less precise and less accurate forecasts, thereby suppressing negative information and inflating market valuations. Notably, our findings highlight that voluntary disclosure decisions reflect a strategic managerial approach, wherein managers weigh the potential benefits of mitigating adverse selection against the associated costs of disclosure. In doing so, we also corroborate the conclusion of Kim, Taylor, and Verrecchia (2021) that these two economic forces – adverse selection and disclosure costs – are jointly determined.

The remainder of the paper is organized as follows: Section 2 reviews relevant literature and develops testing hypotheses, Section 3 presents the data and research design, Section 4 discusses our main empirical results, Section 5 is devoted to additional findings, and Section 6 concludes.

2. Literature review and hypothesis development

2.1. Literature on cybersecurity

The rapid evolution of cyberspace and the escalating costs of cyberattacks in recent decades have motivated a burgeoning body of academic research to investigate the economic consequences of cyber-related issues. This research can be broadly categorized into two main streams based on their temporal focus: *ex-post* and *ex-ante*.

The first stream evaluates the *ex-post* impact of cyberattacks, examining the economic consequences for a firm's valuation after an attack has occurred (e.g., Amir, Levi, & Livne, 2018; Corbet & Gurdgiev, 2019; Tosun, 2021). These studies suggest that firms targeted by cybercrimes experience a substantial decline in stock market returns and an increase in return volatility. Besides, this stream of research explores how firms adjust their investment, governance, and risk management strategies in response to such incidents (e.g., Akey, Lewellen, Liskovich, & Schiller, 2024; Boasiako & Keefe, 2021; Kamiya et al., 2021). However, a significant caveat of these *ex-post* studies arises from their reliance on data from a limited pool of attacked firms, which perhaps is due to the lack of comprehensive measures to capture a firm's inherent cybersecurity risk exposure before an attack occurs. This challenge paves the way for the emergence of the second stream.

The second line of research tackles the issue of data availability by introducing novel *ex-ante* measures for cybersecurity risk. These measures aim to assess a firm's vulnerability to future attacks based on currently available information, overcoming the generalizability limitations of *ex-post* studies. One such example is the work by Florackis et al. (2023). They employ textual analysis to compare a firm's cybersecurity disclosures in the "*Item 1A: Risk Factors*" section of its 10-K filings with disclosures from firms previously compromised by major attacks. This comparison allows them to construct the firm's cyber risk profile based on its disclosure patterns. Their empirical results using this measure reveal that firms with higher *ex-ante* cyber risk exhibit an 8.3% increase in annual returns but also face a greater risk of stock price crashes.

Further supporting these findings, Jiang et al. (2024) utilize machine learning to predict the probability of future cyberattacks based on a combination of financial data and soft information from risk disclosures. Their results echo the presence of a cyber risk premium in the stock returns of firms with higher cyber risk and suggest these firms might underperform when market sentiment towards cybersecurity worsens. Jamilov, Rey, and Tahoun (2023) also contribute to this research area by constructing a text-based measure of cybersecurity risk from quarterly earnings calls. Their work sheds light on both the direct and spill-over effects of cyber risk exposure in the stock market.

Despite the development of various ex-ante measures for cyber risk assessment, research remains relatively scant regarding how managers adapt their firm's policies in response to such risk. One relevant study is Lattanzio and Ma (2023). The authors propose an alternative cyber risk measure based solely on analyzing cyber-related information within 10-K filings, which avoids the constraints associated with comparing to breached firms (as in Florackis et al., 2023). This study finds that firms exposed to cybersecurity risk expedite their innovation cycles by applying for simpler but less valuable patents. Wang, Ho, and Shan (2024) also reach the same conclusions and add that such myopic behavior in innovation arises from managers' lower risk tolerance and their precautionary motives. Our paper builds on this line of literature and enquires into the disclosure tactics managers exploit to manage information flow when dealing with shareholder concerns about cybersecurity.

2.2. Cybersecurity risk and management forecasts

This study questions how *ex-ante* cyber risk affects the issuance of management forecasts, a prominent form of voluntary disclosure closely linked with investors' payoff forecasting needs (Baginski & Rakow, 2012; Hirst et al., 2008). By focusing on management forecasts, we can explore the impact of cyber risk not only on the frequency of forward-looking disclosures but also on the quality of the information convey, such as precision and accuracy.

Nevertheless, we acknowledge that management forecasts are not the only means of voluntary disclosures that can be associated with cybersecurity risk. The association between these two variables could thus be driven by the other narrative form of voluntary disclosures. For instance, managers might opt to provide more detailed narrative descriptions within their filings regarding cybersecurity risk management practices and incident responses. Such actions may serve to mitigate investor concerns about managerial competence and accountability, potentially confounding the relationship between cybersecurity risk and management forecasts. To address this issue, we control for a firm's quality of narrative disclosure as well as past data breach incidents in one of our robustness checks. Also, we utilize alternative voluntary disclosure measures based on the 8-K filings and narrative information from the corporate website to confirm the relationship between cyber risk and voluntary disclosure decisions.

We propose that the presence of cybersecurity risk could induce managers to approach management forecasts in two potential directions. The first possibility is to reduce the frequency of forecasts, given that cybersecurity issues could introduce additional layers of complexity to managers when projecting financial performance, as these issues often involve unquantifiable costs. For instance, the immediate financial costs from a cyber intrusion can vary significantly, ranging from minor disruptions to major losses, contingent on the nature of the attack, the firm's preparedness, and response efficacy. Besides, indirect costs like reputation damage, customer trust erosion, or loss of intellectual property are difficult to estimate and may persist for extended periods (Aldasoro et al., 2022; Kamiya et al., 2021; Makridis, 2021). In addition to these financial uncertainties, cybersecurity risk also requires firms to continuously invest and upgrade their security measures. While these efforts are essential to safeguarding the firm's operations against cyber threats, they can divert funds from other business areas, thereby affecting overall financial performance and crippling managers' ability to forecast accurately (Dewan et al., 2007; Gordon, Loeb, Lucyshyn, & Zhou, 2015; Wang & Alam, 2007).[9]

Furthermore, managers may curtail forecasting frequency due to concerns about proprietary costs associated with disclosure. Prior research indicates that frequent earnings forecasts inadvertently divulge forward-looking and value-relevant information regarding future firm performance, market demand, and operating costs to rival firms, which potentially undermines the firm's competitive advantage (Bamber & Cheon, 1998; Huang et al., 2017; Verrecchia, 1983). Park, Sani, Shroff, and White (2019) additionally emphasize that management forecasts of capital expenditures (capex) can expose significant anticipated cash outlays, affording competitors valuable insights and time to inform strategic responses. Hence, if forecasting firms are concerned about their competitive position, they are less likely to provide these forecasts. This tendency is particularly applicable to firms facing high cyber risk, whose market standing has already been compromised by cybersecurity concerns.

In summary, managers can be reluctant to issue their forecasts due to more costly disclosure processes (Blankespoor et al., 2020; Dorantes, Li, Peters, & Richardson, 2013) and the risk of losing their reputation if they issue low-quality projections (Lee, Matsunaga, & Park, 2012). We refer to this view as the *costly disclosure* hypothesis.

Hypothesis 1a. Managers in firms with higher cybersecurity are less likely to issue management forecasts.

Even though a decision to reduce forecasts may seem appealing in the context of increased forecasting constraints, such a course of action carries its own set of drawbacks. Abstaining from providing management forecasts violates a firm's commitment to transparent disclosure and exacerbates the information asymmetry between management and external stakeholders. Consequently, this can lead to a depreciation of firm value due to an adverse selection problem, wherein potential investors with incomplete knowledge about a firm's information tend to discount its shares (Bertomeu et al., 2022; Diamond & Verrecchia, 1991; Dye, 1985; Kim et al., 2021). Consistent with this notion, Houston et al. (2010) demonstrate that a reduction in management forecasts correlates with decreased analyst coverage and increased analyst forecast errors and dispersion, indicative of diminished forecast precision. Moreover, Chen et al. (2011) document a negative market reaction to the cessation of forecast issuance. Similarly, Baginski and Rakow (2012) reveal a negative relationship between management forecast quality and a firm's cost of equity, implying a lower valuation.

It is plausible to infer that the costs of non-disclosure will be pertinent to firms with higher cybersecurity risks. These firms often operate under increased pressure to enhance their disclosure practices to coordinate greater investor demand for information and more extensive disclosure requirements regarding cybersecurity management practices and data breaches they experience. For instance, the U.S. SEC's 2011 CF Disclosure Guidance (Topic 2) mandates corporate disclosure of significant cybersecurity risks or breaches within 10-K filings. Empirical research, such as that conducted by Jiang et al. (2022), further supports this assertion by showing that firms increase their cybersecurity risk disclosures following data breaches to address the rising information needs of investors, regulators, and other stakeholders. Moreover, Walton et al. (2021), in their literature review, posit that higher exposure to cyber risk is positively related to cybersecurity disclosure. From an experimental perspective, Kelton and Pennington (2020) also reinforce that voluntary

[9] Of note, more advanced cybersecurity measures can impose constraints on the information production process, limiting managers' ability to generate high-quality and timely internal information necessary for accurate forecasts. For instance, data encryption requires decryption before analysis, adding an extra step to the information processing workflow. Similarly, access controls, often implemented through multi-factor authentication, restrict access to sensitive data, potentially slowing down retrieval and analysis for disclosure purposes. Yet, this argument seems minor on the surface relative to the financial and operational uncertainties arising from cyber risk.

cybersecurity disclosures can mitigate investors' contagion effects following a cyber incident. Taken together, one may argue that managers would choose to issue more forecasts to quell investors' concerns about the firm and potentially avoid share undervaluation. We term this view as the *market conditioning* hypothesis.

Hypothesis 1b. Managers in firms with higher cybersecurity are more likely to issue management forecasts.

3. Research design

3.1. Data and sample construction

We construct our main sample by combining data from diverse sources. These include (i) management forecast information from I/B/E/S Guidance, (ii) cybersecurity risk data developed by Lattanzio and Ma (2023) and derived from 10-K filings,[10] (iii) firm fundamentals obtained from Compustat, (iv) stock trading data provided by CRSP, (v) analyst estimations from I/B/E/S Estimates, and (vi) institutional ownership information found in Thomson Reuters/Refinitiv 13F. Only firms with sufficient and non-missing values for the above data fields are included in our sample. The merging of the databases results in a final sample of 51,548 firm-year observations for 6482 unique U.S. firms from 1998 to 2019. This sample period is bounded by the availability of the cybersecurity risk measure.

3.2. Variable definitions

3.2.1. Measure of cybersecurity risk

We opt for Lattanzio and Ma's (2023) cybersecurity score (*CyberScore*) to measure firms' *ex-ante* cybersecurity risk. While other measures like those from Florackis et al. (2023) and Jiang et al. (2024) are worth considering, their data coverage only spans from 2007 to 2018 due to their reliance on past cyberattacks and the mandated risk-factor disclosures in 10-K filings to construct their measures. This timeframe does not align well with the I/B/E/S Guidance database that begins in 1993. Lattanzio and Ma's (2023) measure, with its extensive coverage starting from 1997, provides a better match to management forecast information, allowing us to explore the relationship across a broader scope, leading to more generalizable and representative findings.[11]

CyberScore is constructed based on the textual analysis of 10-K filings using cybersecurity-related terms used by the National Infrastructure Coordinating Council (NICCS) and phrases from Gordon, Loeb, and Sohail (2010).[12] This variable is defined as the natural logarithm of the weighted word count for which the weight of each cybersecurity-related term or phrase is determined by Eq. (1):

$$w_{i,j} = \begin{cases} \dfrac{\left[1+\log\left(tf_{i,j}\right)\right]}{\left[1+\log\left(\alpha_{i,j}\right)\right]} \log\dfrac{N}{df_i} & \text{if } tf_i > 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $w_{i,j}$ represents the weight assigned to each term or phrase i in the jth document; N denotes the total number of 10-K filings throughout the sample period; df_i is the number of documents containing at least one occurrence of the term i; $tf_{i,j}$ is the raw word count i in the jth document, and $\alpha_{i,j}$ is the average word count in the entire document. While the first component of the weight calculation alleviates the impact of high-frequency words with a log transformation, the second component corrects for the commonality of a given term (Loughran & McDonald, 2011).

3.2.2. Measure of voluntary disclosure

Building on the hypothesis that voluntary disclosure represents a managerial response to cybersecurity risk, we use the frequency of management forecasts issued throughout a fiscal year for earnings, sales, and capital expenditures (CapEx) as a measure of disclosure.[13] Extant research emphasizes the critical role of these forecasts in furnishing forward-looking information to the capital market and facilitating the alignment of market expectations with the company's beliefs (Beyer, Cohen, Lys, & Walther, 2010; Healy & Palepu, 2001; Park et al., 2019). Notably, Beyer et al. (2010) demonstrate that management guidance disclosure explains 16% of stock return variation, surpassing the explanatory power of SEC filings (8-Ks, 10-Ks, and press releases), which account for only 4%. Furthermore, research suggests that managers can strategically leverage these forecasts to navigate their firm's information landscape and manage shareholder relations (Abramova, Core, & Sutherland, 2020; Bourveau & Schoenfeld, 2017; Graham, Harvey, & Rajgopal, 2005). Therefore, management forecasts emerge as a suitable and well-supported measure for our investigation.

[10] We are grateful to the authors for kindly sharing their cybersecurity risk data.

[11] As part of our robustness tests, we utilize Florackis et al. (2023) and Jiang et al. (2024) cybersecurity risk proxies as alternative explanatory variables in our regression analyses. The results remain qualitatively unchanged, suggesting that the existing measures of cybersecurity risk share a specific amount of common information.

[12] The final list of terms and phrases used for parsing information from 10-K filings can be found in Appendix B of Lattanzio and Ma (2023).

[13] We incorporate management CapEx forecasts as well, given that a public commitment to a specific investment level can transmit a credible signal of the firm's commitment to significant investment ("skin in the game") and upfront cybersecurity infrastructure enhancements.

We extract management guidance information from the I/B/E/S Guidance database. This feed compiles quantitative company expectations from press releases and corporate event transcripts with earnings forecasts. From the data file, we obtain 309,015 management forecasts issued by our sample firms between 1998 and 2019. We then aggregate these forecasts into the firm-year level using the guidance announcement date and construct our key dependent variable, *Frequency*, as the total number of management forecasts for each firm-year. For those firm-years covered in other datasets that do not appear in the I/B/E/S Guidance database, we set *Frequency* to zero (i.e., no management forecasts issued).[14]

3.2.3. Control variables

We include a standard set of firm-specific control variables frequently used in prior literature (Cheng et al., 2013; Guay, Samuels, & Taylor, 2016). These encompass: (i) *Size* is the natural logarithm of the book value of assets; (ii) *Leverage* is the ratio of short-term plus long-term debt over the book value of assets; (ii) *MTB* equals the market value of equity over the firm's book value; (iv) *Cash* is the ratio of cash and short-term investments to total assets; (v) *R&D* is research and development expenses scaled by lagged total assets; (vi) *CapEx* is capital expenditures divided by lagged total assets; and (vii) *ROA* stands for return-on-assets, computed as the earnings before interest and tax scaled by lagged total assets.

To account for the firm's risk and uncertainty, we include *ROAVol*, the standard deviation of annual return on assets over five years, and *RetVol*, the standard deviation of daily stock returns over the previous fiscal year (Campbell, Cao, Chang, & Chiorean, 2023; Waymire, 1985). Finally, we follow Ali, Klasa, and Yeung (2014) to factor in the company's information environment by including effective bid-ask spread *(Spread)*, analyst coverage (*Analyst)*, and institutional ownership (*IO*). *Spread* is constructed based on Abdi and Ranaldo's (2017) approach using daily high and low prices and closing prices from CRSP. *Analyst* is the natural logarithm of the number of analysts following the firm from I/B/E/S Estimates. *IO* is the proportion of shares held by institutional investors from Thomson Reuters/Refinitiv 13F. The above control variables are lagged by one year relative to *Frequency* (i.e., *t-1*).

3.3. Model specification

To test the association between cybersecurity risk and voluntary disclosure, we specify the following baseline model:

$$Frequency_{i,t} = \alpha + \beta \times CyberScore_{i,t-1} + \gamma' Controls_{i,t-1} + \theta_j + \tau_t + \varepsilon_{i,t} \quad (2)$$

where *i*, *j*, and *t* denote the firm, industry, and year. The dependent variable is *Frequency*. Our main independent variable of interest is $CyberScore_{i,t-1}$, which is the previous-year cybersecurity risk.[15] $Controls_{i,t-1}$ represents the vector of control variables as specified in section 3.2.3. Standard errors of the coefficient estimates are adjusted for heteroskedasticity and clustered at the firm level. Following prior literature (Cho, Kim, & Zang, 2020; Crawford, Huang, Li, & Yang, 2020), we also incorporate industry and year-fixed effects (θ_j and τ_t, respectively) in our baseline specification to tackle the concerns that unobservable heterogeneities across industries and years may drive firms' disclosure decisions. Industry fixed effects are based on two-digit SIC codes.[16] All continuous variables in the regression model are winsorized at the top and bottom 1% of the sample distribution to mitigate the effects of outliers in the data. A detailed explanation of all variables and their sources can be found in Appendix A.

4. Main empirical results

4.1. Descriptive statistics

Panel A of Table 1 shows summary statistics of our main variables separately for the entire sample in Columns (1)–(6), for a subgroup of high-cyber-risk firms (i.e., *CyberScore* is above the sample median) in Columns (7)–(9), and for a subgroup of low-cyber-risk firms (i.e., *CyberScore* is below the sample median) in Columns (10)–(12). Column (13) compares the mean values of voluntary disclosure and its determinants between high- and low-cyber-risk firms.

As can be seen, firms in our sample, on average, release management forecasts roughly six times, with more than half of them doing this at least once per fiscal year. We observe that firms with above-median *CyberScore* values issue their forecasts more frequently (roughly 7.5 times per year) compared to those with below-median *CyberScore* values (around 4.5 times per year), as demonstrated in Column (13). This univariate test lends initial evidence to the notion that companies facing more cybersecurity risk voluntarily disclose

[14] In an untabulated test, we take the natural logarithm of one plus the number of management forecasts to tackle the skewness in management forecast data and use it as an alternative proxy for voluntary disclosure. Our statistical inferences remain qualitatively unchanged.

[15] Using the prior year's *CyberScore* instead of the current year's fits into our research hypothesis that managers utilize their financial forecasts to manage the perception of the market following the disclosure of cybersecurity risk information in the 10-K filings at the end of previous fiscal year. Moreover, given the temporal uncertainty regarding the timing of cybersecurity affairs or disclosures in the 10-Ks, establishing a contemporaneous causal relationship between *CyberScore* and management forecasts is empirically challenging.

[16] Of note, additional tests in section 4.3 indicate that our results are robust to including firm fixed effects (FEs). However, we include only industry fixed effects in our baseline model mainly because both *CyberScore* and *Frequency* remain relatively persistent within firms; therefore, controlling for firm fixed effects can subsume much of the variation in the economic relationship between these two variables (Breuer & DeHaan, 2024). In untabulated regressions of *Frequency* and *CyberScore* on firm and year FEs, we document that firm and year FEs explain 66.55% and 68.27% of the variations in *Frequency* and *CyberScore,* respectively.

Table 1
Descriptive statistics and Pearson correlation matrixPanel A of this table includes descriptive statistics of the key variables of our empirical analyses. The panel presents these statistics separately for our full sample in Columns (1)–(6), a subsample of firms with *CyberScore* values above the sample median in Columns (7)–(9), and a subsample of firms with a subsample of firms with *CyberScore* below the sample median in Columns (10)–(12). Column (13) compares the mean determinants of voluntary disclosure in the two subsamples (high versus low *CyberScore*). Panel B presents Pearson's correlation coefficients between the variables. The full panel sample of U.S. firms contains 51,548 firm-years from 1998 to 2019. All continuous variables are winsorized at the 1% level in both distribution tails. All variable definitions and sources are provided in Appendix A. * indicates significance at the 1% level.

Panel A: Summary statistics

	Full sample						High cyber risk			Low cyber risk			Diff. (7)–(9)
	Obs	Mean	SD	25th	Median	75th	Mean	SD	Median	Mean	SD	Median	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
[1] Frequency	51,548	5.991	8.962	0.000	1.000	9.000	7.485	9.885	3.000	4.498	7.646	0.000	2.988*
[2] CyberScore	51,548	0.058	0.051	0.000	0.059	0.102	0.104	0.026	0.102	0.013	0.020	0.000	0.091*
[3] Size	51,548	6.587	2.087	5.086	6.589	7.969	6.981	2.019	6.946	6.193	2.080	6.178	0.787*
[4] Leverage	51,548	0.213	0.211	0.027	0.167	0.331	0.213	0.208	0.165	0.214	0.214	0.170	−0.001
[5] MTB	51,548	2.777	4.436	1.178	1.892	3.262	2.872	4.464	1.961	2.682	4.405	1.837	0.189*
[6] Cash	51,548	0.188	0.218	0.031	0.096	0.266	0.174	0.197	0.095	0.201	0.236	0.096	−0.026*
[7] R&D	51,548	0.051	0.112	0.000	0.000	0.049	0.040	0.093	0.000	0.062	0.128	0.000	−0.023*
[8] CapEx	51,548	0.047	0.060	0.010	0.028	0.059	0.044	0.054	0.027	0.050	0.065	0.028	−0.005*
[9] ROA	51,548	0.026	0.205	0.008	0.058	0.119	0.047	0.173	0.062	0.005	0.231	0.053	0.041*
[10] ROAVol	51,548	0.091	0.165	0.017	0.040	0.091	0.077	0.153	0.033	0.104	0.175	0.048	−0.027*
[11] RetVol	51,548	0.137	0.173	0.036	0.075	0.163	0.114	0.153	0.060	0.160	0.189	0.092	−0.046*
[12] Spread	51,548	1.355	0.887	0.738	1.093	1.682	1.191	0.790	0.948	1.518	0.946	1.247	−0.328*
[13] Analyst	51,548	1.613	0.966	0.693	1.609	2.398	1.789	0.942	1.946	1.436	0.958	1.386	0.353*
[14] IO	51,548	0.566	0.302	0.315	0.613	0.826	0.621	0.293	0.690	0.511	0.302	0.532	0.111*

Panel B: Pearson correlation matrix

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
[1] Frequency	1.000												
[2] CyberScore	0.188*												
[3] Size	0.219*	0.204*											
[4] Leverage	0.008	−0.021*	0.238*										
[5] MTB	0.083*	0.024*	−0.058*	−0.079*									
[6] Cash	0.001	−0.071*	−0.407*	−0.313*	0.182*								
[7] R&D	−0.038*	−0.115*	−0.408*	−0.141*	0.159*	0.600*							
[8] CapEx	0.039*	−0.055*	−0.020*	0.118*	0.040*	−0.136*	−0.097*						
[9] ROA	0.210*	0.118*	0.377*	0.043*	−0.043*	−0.430*	−0.636*	0.094*					
[10] ROAVol	−0.107*	−0.094*	−0.397*	−0.086*	0.121*	0.443*	0.462*	0.034*	−0.494*				
[11] RetVol	−0.198*	−0.138*	−0.432*	0.032*	−0.004	0.215*	0.309*	−0.008	−0.421*	0.364*			
[12] Spread	−0.270*	−0.191*	−0.585*	−0.015*	−0.052*	0.194*	0.285*	−0.033*	−0.403*	0.331*	0.834*		
[13] Analyst	0.390*	0.202*	0.676*	0.110*	0.101*	−0.088*	−0.103*	0.125*	0.257*	−0.161*	−0.319*	−0.491*	
[14] IO	0.395*	0.193*	0.499*	0.076*	0.039*	−0.095*	−0.179*	0.065*	0.330*	−0.235*	−0.363*	−0.515*	0.610*

information more often. Regarding *CyberScore*, the entire sample has a mean value of 0.058, while those of high- and low-cyber-risk firms are 0.104 and 0.013, respectively. The statistics also suggest that firms with a higher cybersecurity risk tend to be larger, more profitable, and have more analysts following them. Institutional investors hold more shares of these firms than those less exposed to cybersecurity risk.

Lastly, in Panel B of Table 1, we report the Pearson correlation matrix and see no major correlations between *CyberScore* and other covariates, alleviating the concerns about multicollinearity problems in our analysis.[17] More importantly, the correlation coefficient of *CyberScore* with *Frequency* is 0.188 and statistically significant at a 1% level, again in line with our *market conditioning* hypothesis. Combined with the univariate test in Column (13) Panel A, this evidence urges us to explore this effect in a multivariate setting in the subsequent sections.

4.2. Baseline results

Our main hypotheses (Hypotheses 1a & 1b) connote that a company's cybersecurity risk profile influences its managers' willingness to disclose information. This is due to the managers weighing the costs of voluntary disclosure against the benefits of reducing uncertainty about future operations and transparency expectations by their investors. Table 2 presents the estimation results of our baseline regression model, as specified in Eq. (2). The odd-numbered columns depict results without additional variables, whereas the even-numbered columns include a complete set of controls and fixed effects.

In Columns (1) and (2), we estimate Eq. (2) using an OLS estimator, and the results suggest that there is a positive association between *CyberScore* and *Frequency*, statistically significant at a 1% level, indicating that higher cybersecurity risk leads to more disclosure. The impact is economically meaningful. For instance, the coefficient of *CyberScore* in Column (2) is 10.037 (t-stat = 6.07), which implies an annual 8.54% (=10.037 × 0.051/5.991) increase in the frequency of management forecasts associated with a one-standard-deviation increase in cybersecurity risk. Moving to Columns (3) and (4), we document that the same positive relationship remains significant when using *CyberDummy* - an indicator for firms with above-median *CyberScore* values. The estimated coefficient of *CyberDummy* in Column (4) suggests that firms with higher cyber risk issue 12.55% (=0.752/5.991) more forecasts than those with lower risk. Finally, in Columns (5) and (6), we employ Poisson and negative binomial regressions to re-estimate Eq. (2) to account for *Frequency* being a count variable. Our results remain robust to these alternative estimators.

The coefficients of the control variables are also in line with previous studies. For instance, the statistical results suggest that firms disclose more frequently when they are larger, more financially leveraged, hold less cash, have higher market-to-book values, invest more in R&D, and earn higher but less volatile profits, consistent with prior studies (Chen, Li, Lu, & Wang, 2023; Chen et al., 2011; Huang, 2022; Jones, 2007). These firms are also shown to be followed by more financial analysts and have more shares held by institutional investors, implying that these market participants act as an additional source of demand for disclosure in our sample (Abramova et al., 2020; Ajinkya, Bhojraj, & Sengupta, 2005; Basu, Pierce, & Stephan, 2021).

Overall, the results in Table 2 support a robust positive relationship between cybersecurity risk and the frequency of management forecasts, which is in line with Hypothesis 1b. This evidence indicates that managers are more likely to use their forecasts as a tool to manage information flows and investor relations in the context of increased cybersecurity concerns.

4.3. Robustness tests

We perform a battery of robustness tests to validate our baseline findings. These checks include (1) incorporating control variables for narrative disclosure quality, corporate governance characteristics, and managerial incentives to disclose; (2) employing alternative fixed effects specifications to mitigate omitted variable bias; (3) utilizing different measures of cybersecurity risk and alternate proxies for voluntary disclosure; and (4) conducting subsample analyses to evaluate the stability of findings across varying regulatory environments. The results of these tests are available in Table 3. For brevity, we only report the estimated coefficients for *CyberScore* and the variables that are newly introduced to our baseline model in these tests.

4.3.1. Additional controls

Alternative explanations and omitted-variable problems could potentially confound our baseline results. Jiang et al. (2022) find that firms tend to increase their narrative disclosures on cyber risk management after a data breach. Hence, the observed increase in management forecasts perhaps is less of a strategic action by corporate managers, as we proposed, but more of a byproduct of amplified disclosure following a cyber incident. For instance, managers may use earnings guidance to complement narrative disclosures, mitigating investors' concerns about the financial implications of a breach and bolstering beliefs in managerial competence. To address this alternative explanation, we introduce two new control variables capturing the frequency of past cyberattacks (*Data Breach Count*) and firms' narrative disclosure quality (*BOG*) to Eq. (2). *Data Breach Count* is the cumulative number of data breaches that a firm has

[17] We also conducted a (untabulated) variance inflation factor (VIF) test. The results show that *CyberScore* has the lowest VIF (1.08) among our explanatory variables, which confirms that it is not significantly correlated with the other variables.

Table 2
Cybersecurity risk and management forecasts This table presents the estimation results of our baseline Eq. (2) related to the association between cybersecurity risk and the frequency of management forecasts. The regressions are estimated at the firm-year level. The dependent variable is *Frequency*, calculated as a count of management forecasts issued during a given fiscal year *t*. *CyberScore* is the text-based cybersecurity risk measure from Lattanzio and Ma (2023). *CyberDummy* is an indicator variable that equals one for firms with *CyberScore* values higher than the sample median and zero otherwise. Regressions in Columns (1)–(4) are estimated using an OLS estimator, while those in Columns (5) and (6) are based on Poisson and negative binomial (NBReg) estimators, respectively. The control variables include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst,* and *IO*. All these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations. The sample period is from 1998 to 2019.

	Dependent variable = $Frequency_t$					
	OLS Regressions				Poisson	NBReg
Variable	(1)	(2)	(3)	(4)	(5)	(6)
$CyberScore_{t-1}$	32.951***	10.037***			1.921***	2.506***
	(43.36)	(6.07)			(6.97)	(8.37)
$CyberDummy_{t-1}$			2.988***	0.752***		
			(38.38)	(5.39)		
Size		0.156**		0.161**	−0.012	0.001
		(2.02)		(2.08)	(-0.98)	(0.05)
Leverage		0.968**		0.927**	0.168**	0.255***
		(2.41)		(2.30)	(2.41)	(2.89)
MTB		0.054***		0.054***	0.000	0.002
		(4.49)		(4.52)	(0.13)	(0.83)
Cash		−0.857**		−0.881**	−0.277***	−0.252**
		(-1.96)		(-2.01)	(-3.38)	(-2.52)
R&D		3.409***		3.396***	0.809***	0.728***
		(3.98)		(3.96)	(4.02)	(3.46)
CapEx		1.502		1.493	0.432*	0.400
		(1.15)		(1.14)	(1.80)	(1.56)
ROA		4.253***		4.275***	0.973***	1.504***
		(10.60)		(10.63)	(10.55)	(13.93)
ROAVol		−0.940***		−0.925***	−0.334***	−0.183*
		(-3.12)		(-3.06)	(-3.64)	(-1.86)
RetVol		−0.504		−0.469	−0.066	0.002
		(-1.64)		(-1.52)	(-0.63)	(0.01)
Spread		0.007		0.008	−0.138***	−0.128***
		(0.08)		(0.09)	(-4.58)	(-3.87)
Analyst		2.306***		2.328***	0.438***	0.563***
		(21.16)		(21.29)	(19.92)	(21.68)
IO		3.246***		3.260***	0.720***	0.671***
		(9.89)		(9.93)	(12.26)	(10.62)
Obs	51,548	51,548	51,548	51,548	51,548	51,548
Adj./Pseudo R^2	0.035	0.345	0.028	0.344	0.402	0.093
Industry FE	No	Yes	No	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	Yes	Yes

experienced up to a given year *t* based on the information about data breach incidents among U.S. firms from Privacy Rights Clearinghouse.,[1819] *BOG* is the Bog score from Bonsall, Leone, Miller, and Rennekamp (2017), which measures the narrative disclosure quality of 10-K filings.[20] A higher value of *BOG* implies lower readability and lower quality of disclosures. As shown in Column (1), Panel A of Table 3, the *CyberScore* coefficient remains significant at a 1% level after we control for these factors, indicating that our baseline inferences are unlikely to emanate from past cyber incidents and subsequent disclosure changes.

Second, the propensity of managers to issue forecasts and their relationship with cyber risk may be influenced by a firm's internal governance quality (Eng & Mak, 2003; Lawrence, Minutti-Miza, & Vyas, 2018), prevailing financial conditions (Campbell et al., 2023; Healy & Palepu, 2001), and managerial compensation structure (Cheng et al., 2013). As such, we introduce control variables to

[18] The Privacy Rights Clearinghouse (PRC) collect information related to data breach incidents among U.S. firms since 2005. Each record within the PRC database includes the name of the compromised organization and the first disclosure date of the breach. We manually match the breach incidents reported in the PRC database to corresponding company records in Compustat based on company names. This matching procedure yields a dataset consisting of 1442 incidents involving 878 U.S. firms.

[19] For years before 2005, we assume that firms experience no data breach incident. In untabulated results, we also drop these firm-years from our sample and document consistent results.

[20] The Bog scores are publicly available from Professor Brian P. Miller's personal website: https://sites.google.com/iu.edu/professorbrianpmiller/bog-data?authuser=0.

Table 3

Robustness tests This table presents the robustness tests. Panel A reports alternative specifications with different sets of fixed effects and control variables in Columns (1)–(5) and the estimation results using alternative cybersecurity risk measures in Columns (6) and (7). The regressions are estimated at the firm-year level. The dependent variable in this panel is *Frequency*. *CyberScore* is the text-based cybersecurity risk measure from Lattanzio and Ma (2023). *CyberRisk Index* is the cybersecurity risk index from Florackis et al. (2023), and *Cyber risk* is the cyber risk measure from Jiang et al. (2024). *Data Breach Count* is the cumulative total of data breaches a firm has encountered up to a specified year *t*. *BOG* represents the Bog Index - a multifaceted measure of plain English readability developed by Bonsall et al. (2017) - of a firm's 10-K filing. *ICW* is the number of SOX 404 material internal control weaknesses over financial reporting during year *t*. *Board Size* is the number of board members reported in the ISS database. *Board Independence* is the number of independent directors sitting on a board. *E-index* represents the entrenchment index developed by Bebchuk et al. (2009). *Negative News* is the number of quarters in which a firm experienced negative unexpected earnings during a year. *CEO Delta* and *Vega* denote the sensitivity of the CEO's pay to stock price level and volatility, respectively. In Panel B, we replace *Frequency* with alternative disclosure proxies, including the number of management EPS forecasts (*FrequencyEPS*), an indicator variable for the issuance of management forecasts (*Forecast*), the number of 8-K filings with voluntary items and their word count (*Count* and *WCount*), the size of a firm's website and its text size (*SizeW* and *SizeM*), and the number of voluntary non-GAAP disclosures from Bentley et al. (2018) (Non-GAAP). Panel C splits our sample into sub-periods to evaluate whether our baseline results are sensitive to the issuance of (1) SEC ruling over mandatory disclosures of Item 1A - Risk Factors in 2005 and (2) SEC CF Disclosure Guidance: Topic No. 2-Cybersecurity in 2011. *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All of these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations.

Panel A: Alternative specifications and alternative measures of cybersecurity risk

	Dependent variable = Frequency_t						
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CyberScore_{t-1}	9.794***	15.513***	11.506***	2.384**	2.142**		
	(5.86)	(3.99)	(4.17)	(2.57)	(2.02)		
$\text{CyberRisk Index}_{t-1}$						2.264***	
						(4.35)	
Cyber Risk_{t-1}							0.994**
							(2.32)
Data Breach Count	0.231				0.773***		
	(0.56)				(3.75)		
BOG	0.013				0.027		
	(0.97)				(1.27)		
ICW		−0.414***			−0.387***		
		(-3.16)			(-3.70)		
Board Size		−1.721**			−2.330***		
		(-1.99)			(-4.73)		
Board Independence		2.706**			−0.192		
		(1.96)			(-0.24)		
E-index		0.437***			0.095		
		(3.02)			(1.12)		
Negative News			−0.113**		−0.063		
			(-2.55)		(-1.44)		
CEO Delta			−0.000***		0.000*		
			(-3.21)		(1.82)		
CEO Vega			0.001*		0.000		
			(1.71)		(0.16)		
Obs	50,162	14,825	25,668	51,291	13,611	28,313	28,313
Adj. R^2	0.346	0.289	0.320	0.693	0.760	0.335	0.333
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	Yes	Yes	No	No
Industry FE	Yes	Yes	Yes	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	No	No	Yes	Yes
Industry × Year FE	No	No	No	Yes	Yes	No	No
State × Year FE	No	No	No	Yes	Yes	No	No

Panel B: Alternative measures of voluntary disclosure

	FrequencyEPS_t	Forecast_t	Ln(Count)_t	Ln(WCount)_t	Ln(SizeW)_t	Ln(SizeM)_t	Ln(Non-GAAP)_t
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CyberScore_{t-1}	3.253***	3.067***	1.837***	0.360***	2.309***	2.619***	0.234***
	(4.87)	(6.21)	(5.77)	(2.61)	(3.37)	(4.40)	(3.26)
Obs	51,548	51,524	31,441	31,441	35,771	35,436	36,648
Adj./Pseudo R^2	0.274	0.297	0.254	0.188	0.260	0.233	0.260

(*continued on next page*)

Table 3 (*continued*)

Panel B: Alternative measures of voluntary disclosure							
	$FrequencyEPS_t$	$Forecast_t$	$Ln(Count)_t$	$Ln(WCount)_t$	$Ln(SizeW)_t$	$Ln(SizeM)_t$	$Ln(Non\text{-}GAAP)_t$
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: Subsample analyses				
	Dependent variable = $Frequency_t$			
	Pre-2005	Post-2005	Pre-2011	Post-2011
Variable	(1)	(2)	(3)	(4)
$CyberScore_{t-1}$	6.308***	12.453***	9.324***	13.678***
	(4.25)	(5.58)	(5.40)	(4.53)
Obs	17,070	34,478	36,040	15,508
Adj./Pseudo R^2	0.302	0.342	0.338	0.352
Firm controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

account for these potential confounders. Column (2) incorporates proxies for internal governance, which include the number of SOX 404 internal control weaknesses over financial reporting (*ICW*), board size (*Board Size),* and independence (*Board Independence)* along with the entrenchment index (*E-index*) proposed by Bebchuk, Cohen, and Ferrell (2009).[21] Column (3), on the other hand, includes a firm's number of fiscal quarters with negative earnings news during a given year *t* (*Negative News*), and managerial pay sensitivity, *CEO Delta* and *CEO Vega.* After considering all these additional controls, we consistently document positive and statistically significant coefficients of *CyberScore*, further strengthening our results.[22]

4.3.2. Alternative sets of fixed effects

Our main analyses in Table 2 control for industry and year-fixed effects to account for industry- or year-specific attributes that may affect firms' voluntary disclosure decisions. However, recognizing the possibility of omitted variable bias related to firm- or state-level factors, we extend our analysis in Column (4) of Panel A Table 3 by incorporating firm, industry-year, and state-year fixed effects. This comprehensive approach controls for time-invariant firm characteristics and the time-varying industry or state-level institutional factors that might impact the relationship between cybersecurity risk and disclosure.[23] Next, in Column (5), we substantiate the fixed-effect model with all additional controls we discussed in Section 4.3.1. Overall, these changes to the model specification do not materially alter our primary finding of a positive association between cybersecurity risk and voluntary disclosure.

4.3.3. Alternative measures of cybersecurity risk

Other than Lattanzio and Ma (2023), there are two alternative proxies for cyber risk in the existing literature. The first one is the *CyberRisk Index* developed by Florackis et al. (2023), which compares the cybersecurity-related word distribution of the "Item 1A. Risk Factors" in the 10-K filings of training firms with that of a hacked sample and identifies the most similar firms as high cybersecurity risk firms. The second proxy comes from Jiang et al. (2024), who use a variety of machine-learning algorithms to construct a predictive *Cyber risk* score based on firm fundamentals and cybersecurity risk disclosure in "Item 1A. Risk Factors" in the 10-K filings. In Columns (6) and (7) of Panel A, we employ these two alternative measures to confirm that our baseline findings are not sensitive to the choice of cyber risk proxies.[24] Again, the results corroborate a positive relationship between cybersecurity risk and management forecasts.

4.3.4. Alternative measures of voluntary disclosure

Panel B of Table 3 replicates our baseline analysis with diverse voluntary disclosure proxies. In Columns (1) and (2), we find consistent results using the number of management EPS forecasts and an indicator variable for firms issuing at least one management forecast in a given year. Furthermore, we acknowledge that management forecasts may be geared toward conveying financial performance expectations; thus, using them as the sole proxy for voluntary disclosure might underrepresent other channels or types of disclosure that firms might use for this purpose. To address this concern, we consider the following alternative measures: (1) the

[21] Note that the sample size shrinks significantly when we control for corporate governance and managerial pay sensitivity in our regression model due to the coverage of ISS database and Execucomp being limited to S&P1,500 firms in the U.S.

[22] The inclusion of internal control weaknesses as an additional control alleviates a potential measurement error in our cybersecurity risk proxy. Firms that extensively disclose cybersecurity management practices may be assigned higher cybersecurity risk ratings, despite robust security controls. We thank an anonymous referee for suggesting this control.

[23] Because Compustat backfills business addresses, we utilize historical business address from the 10-X Header Data in Software Repository for Accounting and Finance (University of Notre Dame) to define the headquartered state for each firm-year.

[24] We thank the authors for kindly sharing their data with us.

number of 8-K filings with voluntary items and their word counts from He and Plumlee (2020) (denoted as *Count* and *WCount*), (2) the size of a company's website and its size of text elements from Boulland, Bourveau, and Breuer (2021) (denoted as *SizeW* and *SizeM*), and (3) the number of non-GAAP earnings disclosure from Bentley, Christensen, Gee, and Whipple (2018) (denoted as *Non-GAAP*).[25] These alternative proxies are log-transformed to facilitate regression estimation and alleviate skewed distributions. Consistent results across all proxies reinforce our findings.

4.3.5. Subsample analyses

As a final robustness check, we examine the sensitivity of our baseline findings to two major changes in disclosure regulations that could bias the construction of our text-based cybersecurity risk measure. These are (1) the SEC's mandate in 2005 for firms to disclose relevant risk factors in Item 1A of their 10-K filings and (2) the SEC's 2011 ruling requiring disclosure of cybersecurity risks and incidents in Item 1A.[26] We then divide our sample into pre- and post-regulatory change periods and evaluate the consistency of our findings across these subgroups. As shown in Panel C of Table 3, the estimated coefficient of *CyberScore* remains positive and statistically significant at the 1% level across all columns, suggesting that our baseline inferences are not spuriously driven by changes in disclosure regulations.

4.4. Endogeneity tests

Despite comprehensive control variables, potential endogeneity concerns warrant further investigation. For instance, sample selection bias due to the inclusion of firms with available cybersecurity risk data and the potential for omitted variables influencing both disclosure policies and cybersecurity risk may have confounded our results. To address these issues, we employ (1) alternative matching techniques, (2) Heckman's two-step model, (3) instrumental variable estimation, and (4) a difference-in-differences analysis utilizing state-level adoptions of Data Breach Disclosure Laws. A detailed exposition of these endogeneity tests is provided in Online Appendix 1. Overall, all these tests yield consistent findings with the baseline analysis.

5. Additional analyses

5.1. Information demand channel

Hypothesis 1b. suggests that managers use management forecasts to redress increased investor needs for information as cybersecurity risk escalates. Under this premise, the baseline relationship should be more pronounced among firms subject to greater information demand. To explicitly test this assumption, we conduct subsample analyses utilizing three distinct measures of information demand.

The first is the natural logarithm of one plus the number of financial analysts following the company (*Analyst*). This measure follows existing studies highlighting that managers use their disclosures to increase analysts' coverage, which directly affects investors' beliefs and interest in their firms (Graham et al., 2005; Lang & Lundholm, 1996). The second measure is the percentage of institutional ownership (*IO*), which reflects prior studies suggesting institutional investors possess both the incentive and capability to demand greater voluntary disclosure from managers (Ajinkya et al., 2005; Lin, Mao, & Wang, 2018). Finally, we utilize the yearly average of daily EDGAR search volume *(LogSearch*) from the SEC's EDGAR Log data, based on Drake, Roulstone, and Thornock's (2015) procedure, as a proxy for investor information-seeking behavior. This final measure captures the market-based information demand manifested in abnormal investor search activity during the year. Higher values on any of these measures indicate greater information demand.

Next, we split our sample into high- and low-information demand groups based on the median values of the abovementioned proxies. Subsequent subsample analyses, detailed in Table 4, examined how the relationship between cybersecurity risk and management forecasts varied across these groups. To facilitate comparisons, χ^2-statistics for the *CyberScore* coefficient difference between the subsamples are also included. The results indicate a stronger effect of cybersecurity risk on the frequency of management forecasts in firms with higher information demand across all three proxies. For instance, in Columns (1) and (2) of Table 4, the coefficient of the *CyberScore* for high analyst coverage firms (13.242, *t*-stat = 4.75) is more than double that of low coverage firms (5.485, *t*-stat = 4.10). This difference is statistically significant at the 1% level. Collectively, this evidence lends support to the information demand channel.

5.2. Management forecast properties

While we have thus far documented supporting evidence of the *market conditioning* hypothesis, we recognize that this outcome does not necessarily preclude the validity of the *costly disclosure* hypothesis. There is a possibility that managers of firms with higher cybersecurity risk may concurrently face challenges in accurately forecasting financial performance and be incentivized to issue less

[25] We thank Jing He for sharing her 8-K filing voluntary disclosure data with us. The company's website data is accessible at https://github.com/r-boulland/Corporate-Website-Disclosure and the non-GAAP earnings disclosure is publicly available on personal website of Professor Kurt H. Gee at https://sites.google.com/view/kurthgee/data.

[26] We again thank an anonymous referee for suggesting these subsample analyses.

Table 4
Information demand channel This table presents the subsample analyses to examine whether the association between cybersecurity risk and voluntary disclosure is driven by increased information demand from the market participants. The regressions are estimated at the firm-year level. Three proxies are used to gauge investor information demand of a firm in our sample, including the number of financial analysts following the firm (*Analyst*), the proportion of shares held by institutional investors (*IO*), and the number of SEC filings search/downloads from SEC Log File Data (*LogSearch*). The odd-ordered columns report the regression results for a subsample of firms with high (above-median) information demand. In contrast, the even-ordered columns are the estimation results for firms with low (below-median) demand. The dependent variable in this table is *Frequency*. *CyberScore* is the text-based cybersecurity risk measure from Lattanzio and Ma (2023). The control variables include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. χ^2-statistics for the *CyberScore* coefficient difference tests between two subsamples are reported separately for each proxy for information demand. ***,**, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations.

	Dependent variable = $Frequency_t$					
	(1)	(2)	(3)	(4)	(5)	(6)
	High	Low	High	Low	High	Low
Variable	*Information demand = Analyst*		*Information demand = IO*		*Information demand = LogSearch*	
$CyberScore_{t-1}$	13.242***	5.485***	15.455***	6.122***	14.811***	10.714***
	(4.75)	(4.10)	(5.47)	(4.64)	(4.49)	(4.98)
Size	−0.093	0.269***	−0.192	0.397***	−0.319**	0.286***
	(-0.74)	(4.05)	(-1.49)	(5.76)	(-2.48)	(2.77)
Leverage	1.286*	0.527	1.568**	0.215	0.957	0.580
	(1.92)	(1.53)	(2.21)	(0.65)	(1.41)	(1.06)
MTB	0.037*	0.030***	0.027	0.030***	0.047**	0.035*
	(1.86)	(3.34)	(1.21)	(3.21)	(2.34)	(1.89)
Cash	−1.738**	−0.057	−1.109	−0.514	−3.056***	−0.289
	(-1.98)	(-0.16)	(-1.24)	(-1.56)	(-3.60)	(-0.50)
R&D	2.872	2.891***	7.852***	2.043***	5.838***	2.976***
	(1.37)	(4.73)	(3.08)	(3.28)	(3.20)	(2.58)
CapEx	2.308	0.693	2.485	1.433	4.554*	0.959
	(1.08)	(0.64)	(1.07)	(1.40)	(1.70)	(0.49)
ROA	5.923***	2.828***	7.210***	2.481***	5.951***	4.330***
	(6.27)	(10.04)	(7.03)	(8.74)	(7.26)	(8.06)
ROAVol	−2.395***	−0.303	−3.323***	−0.294	−3.142***	0.209
	(-3.64)	(-1.25)	(-3.86)	(-1.24)	(-5.04)	(0.45)
RetVol	−2.624***	−0.052	−1.207	−0.275	0.627	−1.218**
	(-2.88)	(-0.23)	(-1.22)	(-1.18)	(0.78)	(-2.52)
Spread	−1.095***	0.127*	−1.372***	0.154**	−1.195***	0.154
	(-3.34)	(1.95)	(-4.31)	(2.27)	(-4.21)	(1.14)
Analyst	2.273***	1.961***	2.979***	1.702***	2.899***	2.772***
	(7.47)	(19.25)	(15.34)	(17.18)	(14.61)	(18.03)
IO	2.852***	2.880***	4.173***	0.144	2.883***	4.054***
	(5.57)	(8.77)	(3.87)	(0.37)	(5.64)	(8.86)
χ^2 (β[High] - β[Low])	7.12***		10.09***		3.76*	
p-value	0.001		0.002		0.053	
Obs	25,407	26,141	25,774	25,774	18,451	18,463
Adj. R^2	0.320	0.248	0.301	0.257	0.319	0.351
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

precise and positive forecasts to mitigate the adverse market reaction associated with information asymmetry. Such a joint effect of both the *costly disclosure* and *market conditioning* hypotheses is analogous to the findings of Chen et al. (2023), who suggest that managers would rather issue dubious forecasts than cut back their projections when flu outbreaks constrain the information production process. To examine the presence of this joint effect within our sample, we assess how cybersecurity risk impacts the quality of management forecasts reflected through specific attributes such as information content, precision, accuracy, and forecasting horizon.

5.2.1. Good versus bad news

Existing research highlights an asymmetry in managerial disclosure of positive and negative information. Managers may be biased toward withholding negative news to serve their self-interests, such as delaying the adverse impact on stock prices (Kothari, Shu, & Wysocki, 2009), protecting their careers (Baginski et al., 2018; Pae et al., 2016), or maximizing their insider trading profits (Rogers, 2008). Conversely, some studies suggest that firms are more likely to disclose negative but not positive news when facing potential shareholder litigation (Cao & Narayanamoorthy, 2011). Building on these findings, we ponder if the effect of cybersecurity risk on management forecasts established above signifies an asymmetrical managerial disclosure of bad news versus good news to manage investors' expectations.

Our analysis begins by classifying the forecasts. Following Rogers and Stocken (2005), forecasts that meet or exceed analysts'

Table 5
Management forecast properties This table reports how cybersecurity risk affects specific features of management forecasts. All regressions in the table are estimated at the management forecast level (*k*). *Good News* is an indicator variable that equals one for forecasts that meet or exceed financial analysts' consensus estimates and zero otherwise, whereas *Bad News* is set to one for those falling short. *SGood News* is an indicator variable of meet-or-beat forecasts released during months with low (pessimistic) manager sentiment. *SBad News* indicates those short-falling forecasts issued during months with high (optimistic) manager sentiment. *CAR[-1;+1]* is the cumulative daily abnormal return over a three-day window from one day before to one day after the issuance date of management forecasts. The daily abnormal return is the difference between the raw and expected stock returns, estimated based on Fama and French's (1993) three-factor model and the stock return information from day −250 to day −2. *MF Horizon* is the natural logarithm of the number of days between the forecast issuance date and the forecasted fiscal quarter or year-end date. *MF Precision* is the negative of the difference between the upper and lower bounds of the forecast, normalized by the stock price at the time of guidance issuance. Point forecasts, lacking a range, are assigned an *MF Precision* of zero. *MF Accuracy* is calculated as the negative of the absolute difference between a forecasted value (or the lower bound value for a range forecast) and the corresponding actual value at the end of the forecasted period, all scaled by the stock price at the time of guidance issuance. *CyberScore* is the text-based firm-level cybersecurity risk measure from Lattanzio and Ma (2023). *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. χ^2-statistics for the *CyberScore* coefficient difference tests between two subsamples are reported separately for each proxy for information content. ***,**, and * denote the statistical significance of 1%, 5%, and 10%, respectively. *k* and *t* denote management forecast and year, respectively. Obs is the number of observations.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variable	Good News$_{k,t}$	Bad News$_{k,t}$	SGood News$_{k,t}$	SBad News$_{k,t}$	CAR[-1; +1]	MF Horizon$_{k,t}$	MF Precision$_{k,t}$	MF Accuracy$_{k,t}$
CyberScore$_{t-1}$	0.851***	−0.022	0.758***	0.045	0.813**	0.127*	−0.929***	−46.127***
	(3.75)	(-0.09)	(3.53)	(0.18)	(1.97)	(1.79)	(-2.67)	(-3.48)
χ^2 (β[Good] - β[Bad])	4.70**		3.68*					
p-value	0.030		0.055					
Obs	303,033	303,033	268,101	268,101	307,436	281,673	295,748	216,957
Adj./Pseudo R^2	0.0583	0.0329	0.102	0.086	0.008	0.063	0.083	0.103
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

consensus estimates are categorized as good news, while those falling short are classified as bad news. We then define two new indicator variables, *Good News* and *Bad News*, which identify positive and negative forecasts. For each management forecast in our sample, we employ a logit regression model to estimate the impact of the previous-year *CyberScore* and other control variables on the probability of a forecast being categorized as *Good News* or *Bad News*. [27] The results are presented in Columns (1) and (2) of Table 5. Notably, we observe a significant positive coefficient for *CyberScore* in the *Good News* model, while the coefficient for *CyberScore* in the *Bad News* model remains insignificant. A χ^2-test for the difference in the *CyberScore* coefficient reinforces that the effect of *CyberScore* is significantly larger on positive forecasts.[28] The evidence suggests that firms facing higher cyber risk are more likely to issue optimistic, forward-looking guidance than negative forecasts. We interpret this finding as a potential strategic maneuver to manage investor sentiment and market perception where there is less pressure to report good news than bad news in the face of cybersecurity risk.

To address potential concerns regarding the strategic nature of management forecasts, we apply an alternative classification scheme of good and bad news conditioning on the manager sentiment index developed by Jiang, Lee, Martin, and Zhou (2019).[29,30] In particular, we identify *strategic* good news (*SGood news* = 1) as forecasts meeting or exceeding analysts' consensus estimates issued during months with low or pessimistic manager sentiment. On the other hand, *strategic* negative news (*SBad News* = 1) is the forecasts falling short of analysts' estimates issued during months with high or optimistic manager sentiment. We then estimate logit models for *SGood News* and *SBad News* on *CyberScore* in Columns (3) and (4) of Table 5, and the results are consistent with our prior findings.

Lastly, we use the market reactions to the announcement of management forecasts to reaffirm whether the information conveyed through the forecasts is more negative or positive. We estimate the cumulative abnormal stock returns (*CAR*) over the window [−1; +1] centered on the guidance announcement date. The abnormal return is computed as the difference between the raw and expected returns based on Fama and French's (1993) three-factor model using daily stock returns from day −250 to day −2 relative to the

[27] Note that *Good News* and *Bad News* are not perfectly correlated since management forecasts can be qualitative and not attached with an analysts' consensus estimate. These forecasts are coded zero for both *Good News* and *Bad News*.

[28] In Table OA5, we aggregate the management forecast data into firm-years and calculate the percentage of the positive and negative forecasts in the number of management forecasts issued in a given year, denoted as *%Good News* and *%Bad News*, respectively. We then employ regression analysis at the firm level to examine how *CyberScore* and other control variables influence these percentages. The conclusions remain unchanged.

[29] Manager sentiment index is publicly accessible via: https://fuweijiang.weebly.com/research.html.

[30] Our statistical inferences remain unchanged when we use the investor sentiment index by Baker and Wurgler (2006) to adjust the management forecast classification. The results are available upon request.

Table 6
Strategic disclosure and bad news hoarding
This table provides the results of path analyses to examine how increased management forecasts associated with cybersecurity risk affect stock price crash risk. The regressions are estimated at the firm-year level. The firm's future stock price crash risk is proxied by three alternative measures. *NCSKEW* is computed by taking the negative of the third moment of firm-specific weekly returns for each firm during a fiscal year and dividing it by the standard deviation of the returns raised to the third power. *DUVOL* is the log ratio of the standard deviation of the "down" sample to the standard deviation of the "up" sample. *ExtraSigma* is the negative of the worst deviation of firm-specific weekly returns from the average firm-specific weekly return divided by the standard deviation of firm-specific weekly returns. *Frequency* is the number of management forecasts issued in fiscal year *t*. *CyberScore* is the text-based firm-level cybersecurity risk measure from Lattanzio and Ma (2023). *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations.

	Crash Risk$_t$ = NCSKEW$_t$		Crash Risk$_t$ = DUVOL$_t$		Crash Risk$_t$ = ExtraSigma$_t$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Coeff.	Sobel's *p*-value	Coeff.	Sobel's *p*-value	Coeff.	Sobel's *p*-value
Direct Path						
P(CyberScore$_{t-1}$, Crash Risk$_t$)	0.027	0.773	−0.016	0.696	0.022	0.567
Indirect Path						
P(CyberScore$_{t-1}$, Frequency$_t$)	10.221***	0.000	10.218***	0.000	10.209***	0.000
P(Frequency$_t$, Crash Risk$_t$)	0.001**	0.011	0.000*	0.056	0.002***	0.000
P(CyberScore$_{t-1}$, Frequency$_t$) × P(Frequency$_t$, Crash Risk$_t$)	0.015**	0.019	0.005*	0.069	0.025***	0.000
Total Effect	0.042	0.655	−0.011	0.784	0.047	0.581
Mediated % in Total	35.40%		48.70%		52.80%	
Obs	50,437		50,437		50,437	
Firm controls	Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes	
Year FE	Yes		Yes		Yes	

announcement date (with a minimum of 100 non-missing return observations). Results in Column (5) indicate a more positive market reaction to management forecasts issued by firms with higher cybersecurity risk.[31]

5.2.2. Forecasting horizon

Recognizing the strategic nature of management forecasts in the context of cybersecurity risk, we investigate the prevalence of forecasting horizon in such conditions. Existing research suggests two competing perspectives. On the one hand, when firms prioritize meeting investor demand for near-term performance information, managers tend to favor short-term forecasts (Palter, Rehm, & Shih, 2008; Kim et al., 2017). On the other hand, managers facing high costs associated with not issuing forecasts and a commitment to transparency may resort to issuing dubious and long-term forecasts (Chen et al., 2023). The rationale behind this is the extended timeframe before actual results are realized, which allows managers greater leeway to revise erroneous predictions compared to short-term forecasts.

For this test, we estimate *MF Horizon* as the natural logarithm of one plus the number of days between the guidance announcement date and the forecasted fiscal quarter or year-end. An OLS regression analysis in Column (6) of Table 5 examines the relationship between *MF Horizon* and *CyberScore*. The estimated coefficient of *CyberScore* is 0.127 and weakly significant (*t*-stat = 1.79) at a 10% level. Thus, it remains inconclusive whether managers of firms with higher cybersecurity risk are more likely to issue longer-term forecasts. In case this tendency holds, it can arise from a strategic decision to mitigate the potential costs, such as reputational harm, associated with inaccurate short-term forecasts while concurrently satisfying investor demand for information.

5.2.3. Forecasting precision and accuracy

To conclude our analysis of management forecast properties, we assess the impact of cybersecurity risk on the forecasts' quality, focusing on their precision and accuracy. Following Dai, Dai, Ng, and Peng (2023), we construct *MF Precision* as the negative of the difference between the upper and lower bounds of the forecast, normalized by the stock price at the time of guidance issuance. Point forecasts, lacking a range, are assigned an *MF Precision* of zero. Our results in Column (7) reveal a significantly negative coefficient between *CyberScore* and *MF Precision*, suggesting that firms with higher levels of cybersecurity risk issue management forecasts with lower precision.

For forecast accuracy, we define *MF Accuracy* as the negative of the absolute value of the difference between a forecasted value (or the lower bound value for a range forecast) and the corresponding actual value at the end of the forecasted period, all scaled by the

[31] This finding continues to hold when we use the cumulative abnormal stock returns over the window [-3; +3] surrounding the guidance announcement date.

stock price at the time of guidance issuance. In Column (8) of Table 5, we regress *MF Accuracy* on *CyberScore*. Like the precision analysis, we find a statistically significant negative coefficient. This combined evidence, presented in Columns (7) and (8), implicitly indicates the constraints or uncertainties that cybersecurity risk imposes on managers' forecasting processes, as we discussed in Hypothesis 1a.

Taken together, we interpret the empirical evidence in Table 5 as indicative of managers' strategic behavior when navigating the competing demands of increased transparency and daunting forecasting processes caused by cybersecurity risk. More importantly, these results imply that the *market conditioning* and *costly disclosure* hypotheses jointly determine the effects of cyber risk on management forecasts.

5.3. Strategic disclosure and bad news hoarding

5.3.1. Evidence from stock price crash risk

The preceding results reveal that managers engage in disclosing positive news more frequently than negative news when faced with *ex-ante* cybersecurity risk. This asymmetry in information sharing prompts a critical inquiry: Does this selective disclosure represent an attempt to conceal negative information and mitigate market underpricing due to adverse selection problems? This question builds on and extends studies by Florackis et al. (2023) and Obaydin, Xu, and Zurbruegg (2024) that establish a positive relationship between a firm's exposure to cyber risk and its future stock price crashes. We posit that the observed increase in positive managerial forecasts relative to negative ones might be an underlying mechanism that explains the link between cyber risk and stock price crashes.

To test this hypothesis, we employ three alternative crash risk measures. The first measure, *NCSKEW*, is computed by taking the negative of the third moment of firm-specific weekly returns for each firm during a fiscal year and dividing it by the standard deviation of the returns raised to the third power. The second measure, *DUVOL*, focuses on volatility during negative market periods. We separate the weeks in which their returns are above (below) the annual mean and call this the "up" ("down") sample. *DUVOL* is then defined as the log ratio of the standard deviation of the "down" sample to the standard deviation of the "up" sample. Lastly, we compute *ExtraSigma* as the negative of the worst deviation of firm-specific weekly returns from the average firm-specific weekly return divided by the standard deviation of firm-specific weekly returns.

To investigate the indirect effect of cybersecurity risk on crash risk mediated by management forecasts, we conduct path analyses presented in Table 6. We focus on two key coefficients: P(*CyberScore*, *Frequency*) and P(*Frequency*, *Crash Risk*). The first coefficient captures the direct impact of cyber risk on the frequency of management forecasts. The second coefficient, P(*Frequency*, *Crash Risk*), estimates the effect of disclosure frequency on each of the three crash risk variables. Our primary focus lies on the interaction term, P(*CyberScore*, *Frequency*) × P(*Frequency*, *Crash Risk*), reflecting the indirect way cybersecurity risk influences firm-specific stock price crash risk through management forecasts. Across all three proxies, the results suggest that the positive link between cyber risk and crash risk is significantly explained by the tendency of managers to disclose good news more frequently relative to bad news, providing evidence of their bad news hoarding behavior.

5.3.2. Evidence from market reactions to subsequent management forecasts

We provide further evidence of bad news hoarding by examining investor reactions to subsequent forecasts issued by firms with high cybersecurity risk and a history of optimistic projections. We expect that if past positive forecasts were strategically designed to mask negative information, market participants would likely discount the credibility of future forecasts made by these managers.[32] We adopt an empirical design similar to Hutton and Stocken (2021) to test our hypothesis. Specifically, we regress the cumulative abnormal return (*CAR[-1; +1]*) surrounding management forecasts in year t on an interaction term between *CyberScore* and the proportion of positive forecasts released in the preceding year ($t-1$) - *%Good News*. The coefficient on this interaction term quantifies investor responses to subsequent forecasts from firms with elevated cyber risk, contingent upon the degree of their prior optimistic guidance.

Consistent with our hypothesis, the estimation results in Column (1) of Table 7 exhibit a negative and statistically significant coefficient on *CyberScore* × *%Good News*, while the stand-alone coefficients for each of these two variables in the regression are positive. This pattern persists when we substitute *%Good News* with *%SGood News*, which is the proportion of strategic good news based on managers' sentiment index, as discussed in section 5.2.1. These findings evince a market discount applied to future guidance issued by high-cyber-risk firms that have previously engaged in optimistic voluntary disclosure, suggesting that managers of these firms exploit manipulative practices to conceal adverse information.

5.4. Cross-sectional analyses

To strengthen the generalizability of our initial findings, we examine how the association between cybersecurity risk and management forecast varies across firm-level attributes.

5.4.1. Firm reputation

First, we explore the cross-sectional heterogeneity of our baseline relationship within the context of a firm's reputation. Our core

[32] We thank an anonymous referee for proposing this test.

Table 7
Market reactions to subsequent management forecasts This table reports the effects of prior positive management forecasts in firms facing higher cybersecurity risk on market reactions to their subsequent forecasts. The regressions in this table are estimated at the management forecast level (*k*). The dependent variable is *CAR [-1;+1],* which represents the cumulative daily abnormal return over a three-day window from one day before to one day after the issuance date of management forecasts during a given year *t*. The daily abnormal return is the difference between the raw and expected stock returns, estimated based on Fama and French's (1993) three-factor model and the stock return information from day −250 to day −2. *CyberScore* is the text-based firm-level cybersecurity risk measure from Lattanzio and Ma (2023). *%Good News* is the percentage of good news forecasts in the total number of management forecasts issued in the prior year (*t-1*). *%SGood News* is the percentage of strategic good news forecasts in the total number of management forecasts in the prior year (*t-1*). The classification of good news and strategic good news forecasts are discussed in Table 5. *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst,* and *IO*. All these variables are defined in Appendix A. The continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. *k* and *t* denote management forecast and year, respectively. Obs is the number of observations.

	(1)	(2)
Variable	CAR[-1; +1]	CAR[-1; +1]
$CyberScore_{t-1}$	3.282***	1.646***
	(4.54)	(3.07)
$CyberScore_{t-1} \times \%Good\ News_{t-1}$	−4.792***	
	(-3.98)	
$\%Good\ News_{t-1}$	0.035	
	(0.32)	
$CyberScore_{t-1} \times \%SGood\ News_{t-1}$		−2.813**
		(-2.14)
$\%SGood\ News_{t-1}$		0.106
		(0.83)
Obs	307,436	307,436
Adj. R^2	0.007	0.007
Firm controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes

hypothesis posits that management's incentive to issue more forecasts stems from a desire to promote greater legitimacy and a positive outlook in the eyes of their investors. However, the continuous issuance of optimistic forecasts could have unintended consequences, particularly for firms with already well-established market positions or high social standing. In such cases, stakeholders might perceive these forecasts as manipulative tactics prioritizing image over genuine performance, increasing the costs of disclosure to outstep its potential benefits. This logic implies that the influence of cybersecurity risk on management forecasts is attenuated in market-leading firms or those that are well-known and already obtain social reputations.

We test the moderating effect of firm reputation using two new variables. First, following prior literature (Leary & Roberts, 2014), we define *Market Leader* as an indicator for firms ranking in the top tercile of their shares in annual industry sales based on the two-digit SIC codes. The second variable, *High SocialRep*, is an indicator variable that equals one for firms with high social standing. To identify these firms, we first calculate an adjusted corporate social responsibility (CSR) score derived from the KLD database, following the methodology outlined in Deng, Kang, and Low (2013).[33] Firms scoring above the sample median are classified as having high social reputations (*High SocialRep* = 1), with the remainder classified below the median.

In Columns (1) and (2) of Table 8, we augment our baseline model with the interaction terms between these variables and *CyberScore*. The results in Column (2) support our proposition that firms with high social reputations are less likely to engage in excessively optimistic disclosure practices. Meanwhile, the evidence regarding the moderating effect of market leadership on this relationship is inconclusive, as indicated by the weakly significant (at the 10% level) coefficient on the *CyberScore* × *Market Leader* interaction term.

5.4.2. Financial reporting complexity

Another critical aspect to consider is the complexity of information production processes within a firm. Intricate financial information or cumbersome financial reporting procedures can significantly increase the time and effort required for managers to generate forecasts (e.g., Hutton, 2005; Plumlee, 2003). This complexity can weaken the incentives for managers to issue such forecasts as the

[33] Slightly different from Deng et al. (2013), we do not include the corporate governance dimension in our adjusted CSR score to focus specifically on a firm's social and environmental performance.

Table 8
Cross-sectional analyses This table examines the cross-sectional variation in management forecast decisions due to elevated cybersecurity risk. The regressions are estimated at the firm-year level. The dependent variable is *Frequency*. *CyberScore* is the text-based cybersecurity risk measure from Lattanzio and Ma (2023). *Market Leader* is an indicator variable for firms acquiring over 15% of total industry sales in a given year based on the two-digit SIC codes. *High SocialRep* is an indicator variable that equals one for firms scoring above the sample median for CSR performance and zero otherwise. *AMI* is the accounting measurement intensity from Andreicovici et al. (2023). *ARC* is the accounting reporting complexity measure based on the count of XBRL tags disclosed in the 10-K filings in the fiscal year. *Fluidity* is the measure of fluidity measured by Hoberg, Phillips, and Prabhala (2014). *Intangible Intensity* is the natural logarithm of one plus the adjusted-return-on-tangible-assets (ROTA) rank, following Clausen and Hirth (2016). *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All these variables are defined in Appendix A. All continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations.

	Dependent variable = $Frequency_t$					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
$CyberScore_{t-1}$	11.168***	16.931***	17.381***	29.700***	11.787***	17.666***
	(6.12)	(6.10)	(5.44)	(6.78)	(6.66)	(4.91)
$CyberScore_{t-1}$ × Market Leader	−2.326*					
	(-1.75)					
Market Leader	0.137					
	(0.56)					
$CyberScore_{t-1}$ × High SocialRep		−7.276**				
		(-2.13)				
High SocialRep		0.779***				
		(2.78)				
$CyberScore_{t-1}$ × AMI			−1.225***			
			(-4.21)			
AMI			0.053**			
			(2.03)			
$CyberScore_{t-1}$ × ARC				−0.051***		
				(-4.53)		
ARC				0.004***		
				(2.68)		
$CyberScore_{t-1}$ × Fluidity					−0.796**	
					(-2.22)	
Fluidity					−0.000	
					(-0.01)	
$CyberScore_{t-1}$ × Intangible Intensity						−2.471**
						(-2.11)
Intangible Intensity						0.397***
						(5.32)
Obs	51,507	30,282	34,392	17,408	45,675	49,060
Adj. R^2	0.345	0.311	0.325	0.351	0.339	0.345
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

associated disclosure costs become more burdensome. We hypothesize that firms characterized by complex information production processes are less likely to engage in increased disclosure in response to cybersecurity risk.

To test this proposition, we use the following proxies for information production complexity: (i) *accounting measurement intensity* (*AMI*), a text-based measure of a firm's *metering* problems from Andreicovici, van Len, Nikolaev, and Zhang (2023) based on the degree to which the firm uses similar terminology from U.S. GAAP to describe its measurement practices in 10-K filings, and (ii) financial reporting complexity (*ARC)* measured by a count of the XBRL tags for the accounting items disclosed in the 10-K filings developed by Hoitash and Hoitash (2018). We then examine how complexity could condition the association between cyber risk and voluntary disclosure in Columns (3) and (4). We find that the coefficient on *CyberScore* × *AMI* (*ARC*) is significantly negative, suggesting that the impact of cybersecurity risk on voluntary disclosure is lessened for firms with more complex information production processes, as the ongoing cost of issuing forecasts may be higher for such firms.

5.4.3. Proprietary costs of disclosure

Frequent disclosure can incur significant proprietary costs, which arise from the risk of exposing valuable information to competitors and potentially erode a firm's competitive advantage (Verrecchia, 1983). This concern is more pertinent to firms operating in highly competitive markets (Ellis, Fee, & Thomas, 2012) or those with substantial investments in intangible assets, given the non-rivalry nature of these assets (Crouzet, Eberly, Eisfeldt, & Papanikolaou, 2022). Thus, we ask if the intensifying effect of cybersecurity risk on the frequency of management forecasts is weaker under these circumstances.

To test our prediction, we introduce two new variables. First is Hoberg, Phillips, and Prabhala's (2014) fluidity measure (*Fluidity*). This variable captures a firm's exposure to product market threats based on the degree of change observed in competitors' product

Table 9
Moderating mechanisms This table examines the governance role of disciplinary economic actors on the association between cybersecurity risk and management forecasts. The regressions are estimated at the firm-year level. The dependent variable is *Frequency*. *CyberScore* is the text-based cybersecurity risk measure from Lattanzio and Ma (2023). *Hostile Index* is the hostile takeover index developed by Cain et al. (2017). *Exposure to SEC* is the text-based measure of a firm's exposure to the SEC, as defined by Armstrong et al. (2024). *MajorSale* is the percentage of a firm's annual sales made to its major customers, as reported by the Compustat Customer Segments database. *Firm controls* include *Size, Leverage, MTB, Cash, R&D, CapEx, ROA, ROAVol, RetVol, Spread, Analyst*, and *IO*. All these variables are defined in Appendix A. All continuous variables are winsorized at the top and bottom 1% of the sample distribution. Unless otherwise specified, industry and year-fixed effects (FE) are included in the regressions. The *t*-statistics in parentheses are based on heteroskedasticity-consistent and firm-level clustered standard errors. ***, **, and * denote the statistical significance of 1%, 5%, and 10%, respectively. Obs is the number of observations.

	Dependent variable = $Frequency_t$		
Variable	(1)	(2)	(3)
$CyberScore_{t-1}$	14.798***	12.602***	7.705***
	(8.47)	(6.28)	(4.21)
CyberScore x Hostile Index	−26.951***		
	(-2.61)		
Hostile Index	−3.509***		
	(-5.02)		
$CyberScore_{t-1}$ x Exposure to SEC		−7.992**	
		(-2.12)	
Exposure to SEC		−0.198	
		(-0.77)	
$CyberScore_{t-1}$ x MajorSale			9.842***
			(2.78)
MajorSale			−0.690***
			(-2.78)
Obs	51,548	33,952	50,640
Adj. R^2	0.345	0.341	0.345
Firm controls	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

offerings relative to the focal firm's product portfolio. Second is Clausen and Hirth's (2016) intangible intensity measure (*Intangible Intensity*), which ranks firms in our sample using their industry-by-year-adjusted return on tangible assets (=EBITDA/Net PPE).[34] The statistically significant and negative coefficients on the interaction terms between *CyberScore* and these two variables in Columns (5) and (6) support our conjecture. Firms facing greater competitive threats or holding more intangible assets are less likely to engage in frequent disclosure when such disclosures could potentially harm their competitive edge.

5.5. *Moderating mechanisms*

Finally, we inspect what governance mechanisms could effectively moderate the strategic disclosures among managers in firms with elevated cybersecurity risk. We consider the following external disciplinary forces: (i) market for corporate control, which can affect managers' decisions to voluntarily disclose their financial information (Kumar, Langberg, & Sivaramakrishnan, 2012)[35]; (ii) regulatory bodies like the U.S. SEC, given some papers supporting that SEC's oversight influences disclosure practices (Kubick & Lockhart, 2016)[36]; and (iii) major customers, as existing literature suggests that a concentrated customer base can impose greater proprietary costs of voluntary disclosure, disincentivizing managers to disclose their information (Verrecchia & Weber, 2006; Ellis et al., 2012).[37]

The regression results in Columns (1) to (3) of Table 9 indicate that the market for corporate control and regulatory bodies

[34] Clausen and Hirth (2016) posit that firms with higher earnings on their tangible assets likely achieve efficient operational processes, have a highly skilled workforce, advanced technology, and a well-established brand reputation. As such, these firms are more likely to possess a greater wealth of intangible assets.

[35] We use the hostile index from Cain et al. (2017) to examine the disciplinary effect of the market for corporate control on voluntary disclosure. This index differs from other metrics like the *E* or *G*-index by focusing solely on factors beyond a firm's control, such as takeover laws, overall market liquidity, and firm age.

[36] We leverage the measure of firm-level exposure to the U.S. Securities and Exchange Commission (SEC), developed by Armstrong et al. (2024). This measure, denoted as *Exposure to SEC*, is defined as the number of sentences in a firm's 10-K filing that reference the SEC alongside different governmental action words scaled by the total number of sentences in the filing. See Appendix B of Armstrong, Glaese, and Hoopes (2024) for the list of agency action words.

[37] A 'major customer' is defined as one accounting for at least 10% of a firm's total sales revenue, as established by SFAS No. 131. We obtain data on major customers from the Compustat Customer Segments dataset and then calculate *MajorSale*, which reflects the proportion of a firm's total sales generated by its major customers. Firms without major customers have a *MajorSale* value of zero.

potentially act as material countervailing agents, mitigating the increase in optimistic disclosures associated with cybersecurity risk. Conversely, the presence of major customers may amplify strategic management forecasts in firms with higher cybersecurity risk. This finding aligns with previous research (Cao, Hsieh, & Kohlbeck, 2013; Cen, Chen, Hou, & Richardson, 2018; Ma, Wang, Wu, & Zhang, 2020), indicating that firms with a concentrated customer base may be motivated to promptly release favorable information while withholding unfavorable news to preserve relationships with major customers and secure advantageous contract terms.[38]

6. Conclusion

Utilizing a sample of U.S. firms, this study investigates how ex-ante cybersecurity risk affects a firm's voluntary disclosure decisions. Our empirical findings reveal a positive and statistically significant association between cyber risk exposure and the frequency of management forecasts. This economic relationship survives a battery of robustness and identification tests. However, it is important to note that, despite the supportive evidence from endogeneity tests, robustness checks, and heterogeneity analyses, we cannot establish with absolute certainty the causal impact of cybersecurity risk on voluntary disclosure. Our baseline findings, therefore, should be cautiously interpreted.

Additional analyses confirm that managers are inclined to disclose their financial information more frequently to meet the increased investors' demand for information transparency stemming from elevated concerns about the firm's cybersecurity. Interestingly, we find that extended management forecasts associated with cybersecurity risk exhibit characteristics that may limit their credibility and information content. These forecasts tend to present a more positive outlook for firms over longer horizons but exhibit lower precision and accuracy. As such, they may contribute to stock mispricing and potentially predict future stock price crashes.

Our findings hold significant implications for both market participants and regulators. While management forecasts can be a valuable source of information for investors, a degree of caution is warranted, particularly for forecasts issued by firms with higher cybersecurity risk exposure. This awareness can help investors mitigate the risk of being misled by potentially inaccurate (optimistic) disclosures and contribute to a more efficient market.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bar.2025.101578.

Appendix A. Variable definition

Variable name	Description	Data source(s)
Voluntary disclosure proxies		
Frequency	The number of management forecasts issued during a fiscal year *t*	I/B/E/S Guidance
FrequencyEPS	The number of management EPS forecasts issued during a fiscal year *t*	I/B/E/S Guidance
Forecast	An indicator variable that equals one for firms issuing at least one management forecast during a fiscal year *t* and zero otherwise	I/B/E/S Guidance
Ln(Count)	The natural logarithm of one plus the number of 8-K filings that include voluntary items issued by a firm during a fiscal year *t*	He and Plumlee (2020)
Ln(WCount)	The natural logarithm of one plus the word count of 8-K filings that include voluntary items issued by a firm during a fiscal year *t*	He and Plumlee (2020)
Ln(SizeW)	The natural logarithm of the average size of a company's website (in bytes) over the quarters during the fiscal year *t*	Boulland et al. (2021)
Ln(SizeM)	The natural logarithm of the average size of text elements on a company's website (in bytes) over the quarters during the fiscal year *t*	Boulland et al. (2021)
Ln(Non-GAAP)	The natural logarithm of one plus the number of non-GAAP earnings reported during the fiscal year *t*	Kurt H. Gee's website
Cybersecurity risk proxies		
CyberScore	The text-based measure for cybersecurity risk following Lattanzio and Ma's (2023) approach	Lattanzio and Ma (2023)
CyberDummy	An indicator variable that equals one for firms with *CyberScore* values lying above the sample median and zero otherwise	Lattanzio and Ma (2023) and self-constructed
CyberRisk Index	Florackis et al.'s (2023) cybersecurity risk index	Florackis et al. (2023)
Cyber Risk	Jiang et al.'s (2024) cyber risk score	Jiang et al. (2024)
Control variables		

(*continued on next page*)

[38] In untabulated analyses, we also examine the disciplinary role of well-known internal governance mechanisms, including shareholder litigation, board independence, and the design of managerial compensation contracts (i.e., CEO delta and vega). For shareholder litigation risk, we use a dummy variable, *Post UD*, which is set to the value of one for the firm-year observations after the firm's home state adopted the Universal Demand (UD) Laws. These results reveal that none of these internal governance mechanisms works to moderate the positive association between cybersecurity risk and strategic voluntary disclosure.

(*continued*)

Variable name	Description	Data source(s)
Size	The natural logarithm of the book value of total assets	Compustat
Leverage	The ratio of long-term and short-term debt to total assets	Compustat
MTB	The market value of equity scaled by the book value of equity	Compustat
Cash	The ratio of cash and short-term investments to total assets	Compustat
R&D	Research and development expenses scaled by lagged total assets. Missing R&D expenses are set to zero where applicable.	Compustat
CapEx	Capital expenditures scaled by lagged total assets. We set this ratio to zero, where capital expenditures are missing.	Compustat
ROA	Earnings before interest divided by lagged book value of total assets	Compustat
ROAVol	The standard deviation of return-on-assets ratios over the most recent five years.	Compustat
RetVol	The standard deviation of daily stock returns	CRSP
Spread	Effective bid-ask spread, estimated using the model of Abdi and Ranaldo (2017). The spread is computed using the following equations: $Spread = \left(\frac{1}{D}\right)\sum_{d=1}^{N}\sqrt{max(4(c_d - v_d)(c_d - v_{d+1}), 0)}$ and $v_d = \frac{l_d + h_d}{2}$ Where c_d, l_d, and h_d are close, low, and high log prices on day d, respectively. D is the number of trading days in a year.	CRSP & Abdi and Ranaldo (2017)
Analyst	The natural logarithm of the number of analysts following the firm	I/B/E/S Estimates
IO	The proportion of firms' shares held by institutional investors	Thomson/Refinitiv 13F
Data Breach Count	The cumulative number of data breaches a firm has experienced up to a year t	Privacy Rights Clearinghouse
BOG	The Bog index of a firm's 10-K filing during year t	Bonsall et al. (2017)
ICW	The number of SOX 404 material internal control weaknesses over financial reporting during year t.	Audit Analytics
Board Size	The natural logarithm of the number of board members	Institutional Shareholder Services (ISS)
Board Independence	The number of independent directors scaled by the board size	ISS
E-index	Bebchuk, Cohen, and Ferrell's (2009) entrenchment index, constructed based on six entrenchment provisions: staggered boards, poison pills, golden parachutes, supermajority requirements for charter amendments, supermajority requirements for bylaw amendments, and supermajority requirements for mergers.	ISS
Negative News	The number of quarters in which a firm experiences negative unexpected earnings within a given year. Unexpected earnings are computed by subtracting the Earnings Per Share (EPS) excluding extraordinary items of a given quarter (q) from the EPS of the corresponding quarter in the preceding year ($q - 4$), with the result scaled by the share closing price at the end of the quarter (q).	Compustat
CEO Delta	The dollar change in the CEO's wealth for a 1% change in stock price	Execucomp
CEO Vega	The dollar change in the CEO's wealth for a 0.01 change in the standard deviation of stock returns	Execucomp
Other variables for identification strategies in Online Appendix		
Post DBDL	An indicator variable for firm-year observations after a firm's home state adopts Data Breach Disclosure Laws and zero otherwise.	National Conference of State Legislatures
Local CyberScore	The average *CyberScore* value for all firms (excluding focal firms) situated in a specific state.	Compustat, Lattanzio and Ma (2023), 10-X Header Data.
Post	A counterfactual indicator coded based on the corresponding treatment firm's state DBDL adoption	National Conference of State Legislatures
Treat	An indicator variable that equals one for firms located in the states that adopt DBDL a given year and zero for those control firms located in states that have already adopted, not yet adopted, or never adopted DBDL.	National Conference of State Legislatures
Management forecast features		
LogSearch	The yearly average of daily EDGAR search volume from the SEC's EDGAR Log data, based on Drake et al.'s (2015) procedure	SEC's EDGAR Log database
Good News	An indicator variable that equals one for management forecasts that meet or exceed analysts' consensus estimates and zero otherwise.	I/B/E/S Guidance
Bad News	An indicator variable that equals one for management forecasts that fall short of analysts' consensus estimates and zero otherwise.	I/B/E/S Guidance
SGood News	An indicator variable that equals one for management forecasts that meet or exceed analysts' consensus estimates issued during months with low (pessimistic) manager sentiment index.	I/B/E/S Guidance & Jiang et al. (2019)
SBad News	An indicator variable that equals one for management forecasts that fall short of analysts' consensus estimates issued during months with high (optimistic) manager sentiment index.	I/B/E/S Guidance & Jiang et al. (2019)
CAR[-1; +1]	The cumulative abnormal return over the one-day window surrounding the guidance announcement date. The abnormal returns are the difference between raw and expected returns based on Fama and French's (1993) three-factor model using daily stock returns from day −250 to day −2 relative to the announcement date (with a minimum of 100 non-missing return observations).	I/B/E/S Guidance & CRSP
MF Horizon	The natural logarithm of the number of days between the forecast issuance date and the forecasted fiscal quarter or year-end date	I/B/E/S Guidance
MF Precision	The negative of the difference between the upper and lower bounds of the forecast, normalized by the stock price at the time of guidance issuance. Point forecasts are assigned a value of zero.	I/B/E/S Guidance
MF Accuracy	The negative of the absolute difference between a forecasted value (or the lower bound value for a range forecast) and the corresponding actual value at the end of the forecasted period, all scaled by the stock price at the time of guidance issuance.	I/B/E/S Guidance
Crash risk measures		
NCSKEW	The negative of the third moment of firm-specific weekly returns for each firm during a fiscal year, divided by the standard deviation of the returns raised to the third power.	CSRP

(*continued on next page*)

(*continued*)

Variable name	Description	Data source(s)
DUVOL	The log ratio of the standard deviation of the "down" sample to the standard deviation of the "up" sample. The "up" ("down") weeks are those with returns above (below) the annual average return.	CRSP
ExtraSigma	The negative of the worst deviation of firm-specific weekly returns from the average firm-specific weekly return divided by the standard deviation of firm-specific weekly returns.	CRSP
Other variables		
%Good News	The percentage of good news forecasts in the total number of management forecasts issued during a fiscal year.	I/B/E/S Guidance
%SGood News	The percentage of strategic good news forecasts in the total number of management forecasts issued during a fiscal year.	I/B/E/S Guidance & Jiang et al. (2019)
Market Leader	An indicator variable for firms acquiring more than 15 percent of their industry's total sales and zero otherwise	Compustat
High SocialRep	An indicator variable for firms with above-median adjusted CSR score, estimated following Deng, Kang, and Low's (2013) approach	MSCI KLD
AMI	The accounting measurement intensity developed by Andreicovici et al. (2023) based on the textual similarity between the terminologies used in firms' 10-K filings and those used in U.S. GAAP pronouncements.	Andreicovici et al. (2023)
ARC	The financial reporting complexity, measured by a count of the XBRL tags for the accounting items disclosed in the 10-K filings developed by Hoitash and Hoitash (2018)	Hoitash and Hoitash (2018)
Fluidity	Hoberg et al.'s (2014) fluidity measure	Hoberg-Phillips Data Library
Intangible Intensity	The natural logarithm of one plus the adjusted-ROTA rank measure, where ROTA represents the return on tangible assets (EBITDA/Net PPE) adjusted for the industry-year median and then normalized by the industry-year standard deviation.	Compustat, Clausen and Hirth (2016)
Hostile Index	The hostile takeover index from Cain, McKeon, and Solomon (2017)	Cain et al. (2017)
Exposure to SEC	The firm-level exposure to the U.S. SEC, calculated as the number of sentences in a firm's 10-K filing in a given year that reference the SEC along with agency action words divided by the total number of sentences.	Armstrong, Glaeser, and Hoopes (2024)
MajorSale	The percentage of a firm's annual sales made to its major customers	Compustat Customer Segments database

Data availability

Data will be made available on request.

References

Abdi, F., & Ranaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies, 30*(12), 4437–4480.
Abramova, I., Core, J. E., & Sutherland, A. (2020). Institutional investor attention and firm disclosure. *The Accounting Review, 95*(6), 1–21.
Ajinkya, B., Bhojraj, S., & Sengupta, P. (2005). The association between outside directors, institutional investors, and the properties of management earnings forecasts. *Journal of Accounting Research, 43*(3), 343–376.
Akey, P., Lewellen, S., Liskovich, I., & Schiller, C. (2024). *Hacking corporate reputations. ECGI working paper series in finance*.
Aldasoro, I., Gambacorta, L., Giudici, P., & Leach, T. (2022). The drivers of cyber risk. *Journal of Financial Stability, 60*, Article 100898.
Ali, A., Klasa, S., & Yeung, E. (2014). Industry concentration and corporate disclosure policy. *Journal of Accounting and Economics, 58*(2–3), 240–264.
Amir, E., Levi, S., & Livne, T. (2018). Do firms underreport information on cyber-attacks? Evidence from capital markets. *Review of Accounting Studies, 23*(3), 1177–1206.
Andreicovici, I., van Len, L., Nikolaev, V. V., & Zhang, R. (2023). Accounting measurement intensity. *TRR266 Accounting for Transparency Working Paper Series No. 30*.
Armstrong, D. M., Glaeser, S., & Hoopes, J. L. (2024). Measuring firm exposure to government agencies. *Journal of Accounting and Economics*.
Baginski, S. P., Campbell, J. L., Hinson, L. A., & Koo, D. S. (2018). Do career concerns affect the delay of bad news disclosure? *The Accounting Review, 93*(2), 61–95.
Baginski, S. P., & Rakow, K. C. (2012). Management earnings forecast disclosure policy and the cost of equity capital. *Review of Accounting Studies, 17*(2), 279–321.
Baker, M., & Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance, 61*(4), 1645–1680.
Bamber, L. S., & Cheon, Y. S. (1998). Discretionary management earnings forecast disclosures: Antecedents and outcomes associated with forecast venue and forecast specificity choices. *Journal of Accounting Research, 36*(2), 167–190.
Basu, R., Pierce, S. R., & Stephan, A. (2021). *The effect of investor inattention on non-GAAP disclosure*. Working Paper.
Bebchuk, L., Cohen, A., & Ferrell, A. (2009). What matters in corporate governance. *Review of Financial Studies, 22*(2), 783–827.
Bentley, J. W., Christensen, T. E., Gee, K. H., & Whipple, B. C. (2018). Disentangling managers' and analysts' non-GAAP reporting. *Journal of Accounting Research, 56*(4), 1039–1081.
Berkman, H., Jona, J., Lee, G., & Soderstrom, N. (2018). Cybersecurity awareness and market valuations. *Journal of Accounting and Public Policy, 37*(6), 508–526.
Bertomeu, J., Marinovic, I., Terry, S. J., & Varas, F. (2022). The dynamics of concealment. *Journal of Financial Economics, 143*(1), 227–246.
Beyer, A., Cohen, D. A., Lys, T. Z., & Walther, B. R. (2010). The financial reporting environment: Review of the recent literature. *Journal of Accounting and Economics, 50*(2–3), 296–343.
Blankespoor, E., deHaan, E., & Marinovic, I. (2020). Disclosure processing costs, investors' information choice, and equity market outcomes: A review. *Journal of Accounting and Economics, 70*(2–3).
Boasiako, K. A., & Keefe, M. O. C. (2021). Data breaches and corporate liquidity management. *European Financial Management, 27*(3), 528–551.
Bonsall, S. B., Leone, A. J., Miller, B. P., & Rennekamp, K. (2017). A plain English measure of financial reporting readability. *Journal of Accounting and Economics, 63*(2–3), 329–357.
Boulland, R., Bourveau, T., & Breuer, M. (2021). Corporate websites: A new measure of voluntary disclosure. *SSRN Working Paper*.
Bourveau, T., & Schoenfeld, J. (2017). Shareholder activism and voluntary disclosure. *Review of Accounting Studies, 22*(3), 1307–1339.
Breuer, M., & DeHaan, E. (2024). Using and interpreting fixed effects models. *Journal of Accounting Research, 62*(4), 1183–1226.
Brockman, P., Khurana, I. K., & Martin, X. (2008). Voluntary disclosures around share repurchases. *Journal of Financial Economics, 89*(1), 175–191.

BusinessWire. (2021). Cyber warfare: Report on 2020 shows triple-digit increases across all Malware types. Available at: https://www.businesswire.com/news/home/20210210005312/en/Cyber-Warfare-Report-on-2020-Shows-Triple-Digit-Increases-Across-All-Malware-Types.
Cain, M. D., McKeon, S. B., & Solomon, S. D. (2017). Do takeover laws matter? Evidence from five decades of hostile takeovers. *Journal of Financial Economics, 124*(3), 464–485.
Campbell, J. L., Cao, S. S., Chang, H. S., & Chiorean, R. (2023). The implications of firms' derivative usage on the frequency and usefulness of management earnings forecasts. *Contemporary Accounting Research, 40*(4), 2409–2445.
Cao, J., Hsieh, H.-Y., & Kohlbeck, M. J. (2013). Do major customers influence voluntary disclosure? *SSRN Working Paper*.
Cao, Z., & Narayanamoorthy, G. S. (2011). The effect of litigation risk on management earnings forecasts. *Contemporary Accounting Research, 28*(1), 125–173.
Cen, L., Chen, F., Hou, Y., & Richardson, G. (2018). Strategic disclosures of litigation loss contingencies: When customer-supplier relationships are at risk. *The Accounting Review, 93*(2), 137–159.
Chen, C., Li, L. L., Lu, L. Y., & Wang, R. (2023). Flu fallout: Information production constraints and corporate disclosure. *Journal of Accounting Research, 61*(4), 1063–1108.
Chen, S., Matsumoto, D., & Rajgopal, S. (2011). Is silence golden? An empirical analysis of firms that stop giving quarterly earnings guidance. *Journal of Accounting and Economics, 51*(1–2), 134–150.
Cheng, Q., Luo, T., & Yue, H. (2013). Managerial incentives and management forecast precision. *The Accounting Review, 88*(5), 1575–1602.
Cho, Y. J., Kim, Y., & Zang, Y. (2020). Information externalities and voluntary disclosure: Evidence from a major customer's earnings announcement. *The Accounting Review, 95*(6), 73–96.
Clausen, S., & Hirth, S. (2016). Measuring the value of intangibles. *Journal of Corporate Finance, 40*, 110–127.
Corbet, S., & Gurdgiev, C. (2019). What the hack: Systematic risk contagion from cyber events. *International Review of Financial Analysis, 65*, Article 101386.
Crawford, S., Huang, Y., Li, N., & Yang, Z. (2020). Customer concentration and public disclosure: Evidence from management earnings and sales forecasts. *Contemporary Accounting Research, 37*(1), 131–159.
Crouzet, N., Eberly, J. C., Eisfeldt, A. L., & Papanikolaou, D. (2022). The economics of intangible capital. *The Journal of Economic Perspectives, 36*(3), 29–52.
Dai, L., Dai, R., Ng, L., & Peng, Z. (2023). Global outsourcing and voluntary disclosure. *Journal of Business Finance & Accounting*, 1–34. December 2022.
Deng, X., Kang, J. K., & Low, B. S. (2013). Corporate social responsibility and stakeholder value maximization: Evidence from mergers. *Journal of Financial Economics, 110*(1), 87–109.
Dewan, S., Shi, C., & Gurbaxani, V. (2007). Investigating the risk-return relationship of information technology investment: Firm-level empirical analysis. *Management Science, 53*(12), iv–1989.
Diamond, D. W., & Verrecchia, R. E. (1991). Disclosure, liquidity, and the cost of capital. *The Journal of Finance, 46*(4), 1325–1359.
Dorantes, C. A., Li, C., Peters, G. F., & Richardson, V. J. (2013). The effect of enterprise systems implementation on the firm information environment. *Contemporary Accounting Research, 30*(4), 1427–1461.
Drake, M. S., Roulstone, D. T., & Thornock, J. R. (2015). The determinants and consequences of information acquisition via EDGAR. *Contemporary Accounting Research, 32*(3), 1128–1161.
Dye, R. A. (1985). Disclosure of nonproprietary information. *Journal of Accounting Research, 23*(1), 123.
Ellis, J. A., Fee, C. E., & Thomas, S. E. (2012). Proprietary costs and the disclosure of information about customers. *Journal of Accounting Research, 50*(3), 685–727.
Eng, L. L., & Mak, Y. T. (2003). Corporate governance and voluntary disclosure. *Journal of Accounting and Public Policy, 22*(4), 325–345.
Ettredge, M. L., Guo, F., & Li, Y. (2018). Trade secrets and cybersecurity breaches. *Journal of Accounting and Public Policy, 37*(6), 564–585.
Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics, 33*(1), 3–56.
Feng, M., Li, C., & McVay, S. (2009). Internal control and management guidance. *Journal of Accounting and Economics, 48*(2–3), 190–209.
Florackis, C., Louca, C., Michaely, R., & Weber, M. (2023). Cybersecurity risk. *Review of Financial Studies, 36*(1), 351–407.
Gong, G., Li, L. Y., & Xie, H. (2009). The association between management earnings forecast errors and accruals. *The Accounting Review, 84*(2), 497–530.
Gordon, L. A., Loeb, M. P., Lucyshyn, W., & Zhou, L. (2015). Increasing cybersecurity investments in private sector firms. *Journal of Cybersecurity, 1*(1), 3–17.
Gordon, L. A., Loeb, M. P., & Sohail, T. (2010). Market value of voluntary disclosures concerning information security. *MIS Quarterly, 34*(3), 567–594.
Graham, J. R., Harvey, C. R., & Rajgopal, S. (2005). The economic implications of corporate financial reporting. *Journal of Accounting and Economics, 40*(1–3), 3–73.
Guay, W., Samuels, D., & Taylor, D. (2016). Guiding through the Fog: Financial statement complexity and voluntary disclosure. *Journal of Accounting and Economics, 62*(2–3), 234–269.
He, J., & Plumlee, M. A. (2020). Measuring disclosure using 8-K filings. *Review of Accounting Studies, 25*(3), 903–962.
Healy, P. M., & Palepu, K. G. (2001). Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics, 31*(1–3), 405–440.
Hirst, D. E., Koonce, L., & Venkataraman, S. (2008). Management earnings forecasts: A review and framework. *Accounting Horizons, 22*(3), 315–338.
Hoberg, G., Phillips, G., & Prabhala, N. (2014). Product market threats, payouts, and financial flexibility. *The Journal of Finance, 69*(1), 293–324.
Hoitash, R., & Hoitash, U. (2018). Measuring accounting reporting complexity with XBRL. *The Accounting Review, 93*(1), 259–287.
Houston, J. F., Lev, B., & Tucker, J. W. (2010). To guide or not to guide? Causes and consequences of stopping quarterly earnings guidance. *Contemporary Accounting Research, 27*(1), 143–185.
Huang, Y. (2022). Government subsidies and corporate disclosure. *Journal of Accounting and Economics, 74*(1), Article 101480.
Huang, Y., Jennings, R., & Yu, Y. (2017). Product market competition and managerial disclosure of earnings forecasts: Evidence from important tariff rate reductions. *The Accounting Review, 92*(3), 185–207.
Hutton, A. P. (2005). Determinants of managerial earnings guidance prior to regulation fair disclosure and bias in analysts' earnings forecasts. *Contemporary Accounting Research, 22*(4), 867–914.
Hutton, A. P., & Stocken, P. C. (2021). Prior forecasting accuracy and investor reaction to management earnings forecasts. *Journal of Financial Reporting, 6*(1), 87–107.
Jamilov, R., Rey, H., & Tahoun, A. (2023). *The anatomy of cyber risk*. Working Paper.
Jiang, H., Khanna, N., Yang, Q., & Zhou, J. (2024). The cyber risk premium. *Management Science*.
Jiang, F., Lee, J. A., Martin, X., & Zhou, G. (2019). Management sentiment and stock returns. *Journal of Financial Economics, 132*(1), 126–149.
Jiang, W., Legoria, J., Reichelt, K. J., & Walton, S. (2022). Firm use of cybersecurity risk disclosures. *Journal of Information Systems, 36*(1), 151–180.
Jones, D. A. (2007). Voluntary disclosure in R&D-intensive industries. *Contemporary Accounting Research, 24*(2), 489–522.
Jung, W. O., & Kwon, Y. (1988). Disclosure when the market is unsure of information endowment of managers. *Journal of Accounting Research, 26*(1), 146–153.
Kamiya, S., Kang, J. K., Kim, J., Milidonis, A., & Stulz, R. M. (2021). Risk management, firm reputation, and the impact of successful cyberattacks on target firms. *Journal of Financial Economics, 139*(3), 719–749.
Kelton, A. S., & Pennington, R. R. (2020). Do voluntary disclosures mitigate the cybersecurity breach contagion effect? *Journal of Information Systems, 34*(3), 133–157.
Kim, Y., Su, L., Nancy, Zhu, X., & Kevin. (2017). Does the cessation of quarterly earnings guidance reduce investors' short-termism? *Review of Accounting Studies, 22*(2), 715–752.
Kim, J. M., Taylor, D. J., & Verrecchia, R. E. (2021). Voluntary disclosure when private information and disclosure costs are jointly determined. *Review of Accounting Studies, 26*(3), 971–1001.
Koller, T., Relm, W., & Williams, Z. (2021). *Should you start issuing EPS guidance again?* McKinsey & Company. https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/should-you-start-issuing-eps-guidance-again.
Kothari, S. P., Shu, S., & Wysocki, P. D. (2009). Do managers withhold bad news. *Journal of Accounting Research, 47*(1), 241–276.
Kubick, T. R., & Lockhart, G. B. (2016). Proximity to the SEC and stock price crash risk. *Financial Management, 45*(2), 341–367.
Kumar, P., Langberg, N., & Sivaramakrishnan, K. (2012). Voluntary disclosures, corporate control, and investment. *Journal of Accounting Research, 50*(4), 1041–1076.
Lang, M. H., & Lundholm, R. J. (1996). Corporate disclosure policy and analyst behavior. *The Accounting Review, 71*(4), 467–492.
Lattanzio, G., & Ma, Y. (2023). Cybersecurity risk and corporate innovation. *Journal of Corporate Finance, 82*, Article 102445.

Lawrence, A., Minutti-Miza, M., & Vyas, D. (2018). Is operational control risk informative of financial reporting deficiencies? *Auditing: A Journal of Practice & Theory, 37*(1), 139–165.
Leary, M. T., & Roberts, M. R. (2014). Do peer firms affect corporate financial policy? *The Journal of Finance, 69*(1), 139–178.
Lee, S., Matsunaga, S. R., & Park, C. W. (2012). Management forecast accuracy and CEO turnover. *The Accounting Review, 87*(6), 2095–2122.
Leuz, C., & Verrecchia, R. E. (2000). The economic consequences of increased disclosure. *Journal of Accounting Research, 38*, 91–124, 2000.
Lin, Y., Mao, Y., & Wang, Z. (2018). Institutional ownership, peer pressure, and voluntary disclosures. *The Accounting Review, 93*(4), 283–308.
Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-ks. *The Journal of Finance, 66*(1), 35–65.
Ma, X., Wang, W., Wu, J., & Zhang, W. (2020). Corporate customer concentration and stock price crash risk. *Journal of Banking & Finance, 119*, Article 105903.
Makridis, C. A. (2021). Do data breaches damage reputation? Evidence from 45 companies between 2002 and 2018. *Journal of Cybersecurity, 7*(1).
McAfee. (2020). New McAfee report estimates global cybercrime losses to exceed $1 trillion. Available at: https://www.mcafee.com/de-ch/consumer-corporate/newsroom/press-releases/press-release.html?news_id=6859bd8c-9304-4147-bdab-32b35457e629.
Obaydin, I., Xu, L., & Zurbruegg, R. (2024). The unintended cost of data breach notification laws: Evidence from managerial bad news hoarding. *Journal of Business Finance & Accounting*.
Pae, S., Song, C. J., & Yi, A. C. (2016). Career concerns and management earnings guidance. *Contemporary Accounting Research, 33*(3), 1172–1198.
Palter, R. N., Rehm, W., & Shih, J. (2008). Communicating with the right investors. *McKinsey Quarterly, 2*, 64.
Park, J., Sani, J., Shroff, N., & White, H. (2019). Disclosure incentives when competing firms have common ownership. *Journal of Accounting and Economics, 67*(2–3), 387–415.
Pham, M. D. M., & Nguyen, T. H. (2024). When a gift resembles a trojan horse: CEO stock gift and stock price crash risk. *The British Accounting Review, 56*(4), Article 101235.
Plumlee, M. A. (2003). The effect of information complexity on analysts' use of that information. *The Accounting Review, 78*(1), 275–296.
PwC. (2018). Investors put cybersecurity top of the business threat list. Available at: https://www.pwc.com/kz/en/pwc-news/what-new/investors-put-cybersecurity.html.
Rogers, J. L. (2008). Disclosure quality and management trading incentives. *Journal of Accounting Research, 46*(5), 1265–1296.
Rogers, J. L., & Stocken, P. C. (2005). Credibility of management forecasts. *The Accounting Review, 80*(4), 1233–1260.
Shroff, N., Sun, A. X., White, H. D., & Zhang, W. (2013). Voluntary disclosure and information asymmetry: Evidence from the 2005 securities offering reform. *Journal of Accounting Research, 51*(5), 1299–1345.
Tang, M., Minye, Yao, L., & Zarowin, P. (2023). Once is not enough: An analysis of management updates of annual earnings forecasts. *SSRN Working Paper*.
Tosun, O. K. (2021). Cyber-attacks and stock market activity. *International Review of Financial Analysis, 76*(April), Article 101795.
Verrecchia, R. E. (1983). Discretionary disclosure. *Journal of Accounting and Economics, 5*, 179–194.
Verrecchia, R. E., & Weber, J. (2006). Redacted disclosure. *Journal of Accounting Research, 44*(4), 791–814.
Violino, B. (2023). *Phishing attacks are increasing and getting more sophisticated. Here's how to avoid them*. CNBC Cyber Report. https://www.cnbc.com/2023/01/07/phishing-attacks-are-increasing-and-getting-more-sophisticated.html.
Walton, S., Wheeler, P. R., Zhang, Y., & Zhao, X. (2021). An integrative review and analysis of cybersecurity breach: Current state and future directions. *Journal of Information Systems, 35*(1), 155–186.
Wang, L., & Alam, P. (2007). Information technology capability: Firm valuation, earnings uncertainty, and forecast accuracy. *Journal of Information Systems, 21*(2), 27–48.
Wang, J., Ho, C. Y., Chloe, & Shan, Y. G. (2024). Does cybersecurity risk stifle corporate innovation activities? *International Review of Financial Analysis, 91*(October), Article 103028.
Waymire, G. (1985). Earnings volatility and voluntary management forecast disclosure. *Journal of Accounting Research, 23*(1), 268–295.
World Economic Forum. (2020). The global risks report. Available at: https://www3.weforum.org/docs/WEF_Global_Risk_Report_2020.pdf.